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CR23532 – CYBER AND DIGITAL FORENSICS RECORD

Submitted by

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in partial fulfillment of the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

ACADEMIC YEAR: 2026 - 2027

INDEX

S.NO	TABLE OF CONTENTS	PAGE NUMBER
1	INTRO TO DIGITAL FORENSICS	1
2	STUDY OF COMPUTER FORENSICS AND DIGITAL FORENSICS TOOLS USED FOR FORENSIC INVESTIGATION	6
3	WINDOWS FORENSICS 1	11
4	LIVE FORENSIC INVESTIGATION USING AUTOPSY	19
5	FTK IMAGER - DISK IMAGING AND EVIDENCE ACQUISITION	30
6	FIND LAST CONNECTED USB ON YOUR SYSTEM	38
7	HOW TO VIEW LAST ACTIVITY OF YOUR PC	42
8	HOW TO EXTRACT BROWSER ARTIFACTS	47
9	COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE	51
10	HOW TO COLLECT EMAIL EVIDENCE IN VICTIM PC	56
11	HIDE AND EXTRACT ANY TEXT FILE BEHIND AN IMAGE FILE USING COMMAND PROMPT	60
12	HOW TO EXTRACT EXCHANGEABLE IMAGE FILE FORMAT (EXIF) DATA FROM IMAGE FILES USING EXIFREADER SOFTWARE	64

EXPERIMENT NO: 1 **DATE: 16/07/26**

INTRO TO DIGITAL FORENSICS

AIM:

To learn about digital forensics and related processes and experiment with a practical example.

PROCEDURE:

1. Navigate to the TryHackMe room 'Intro to Digital Forensics'.
2. Identify the digital evidence from the crime scene photo.
3. Learn forensic phases.
4. Inspect document metadata to find the document author.
5. Extract EXIF metadata to find the camera model.
6. Read GPS metadata coordinates to pinpoint the location.

TASKS:



Figure 1.1 TryHackMe room completed

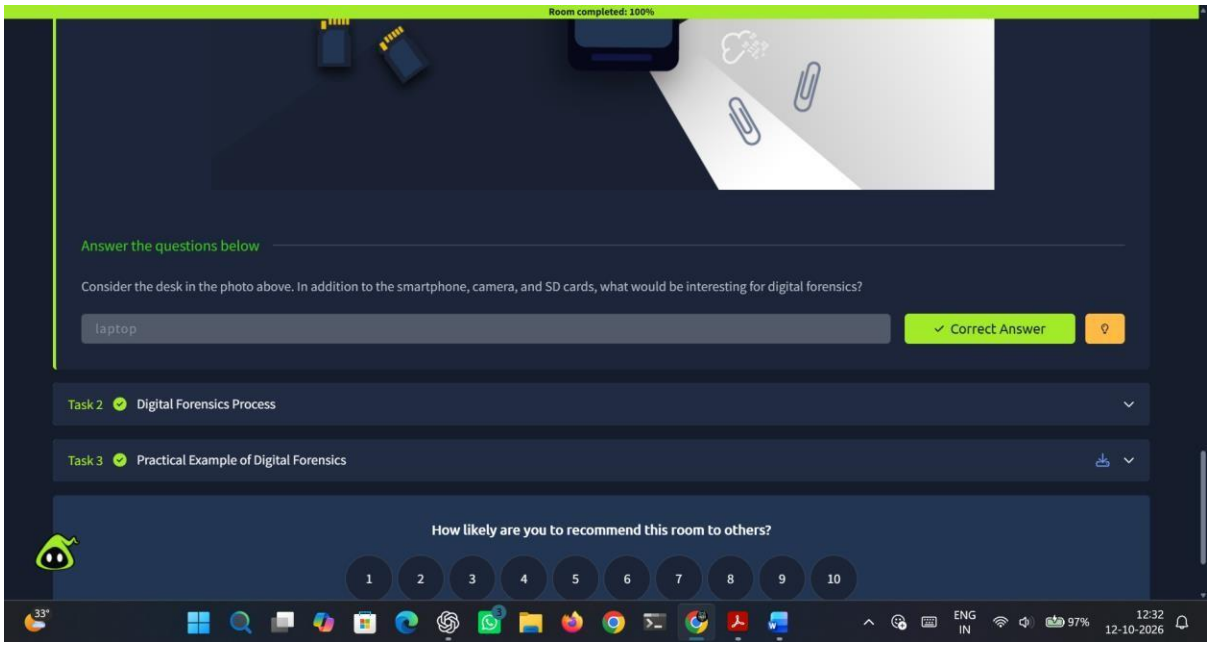


Figure 1.2 Task 1: Introduction to Digital Forensics

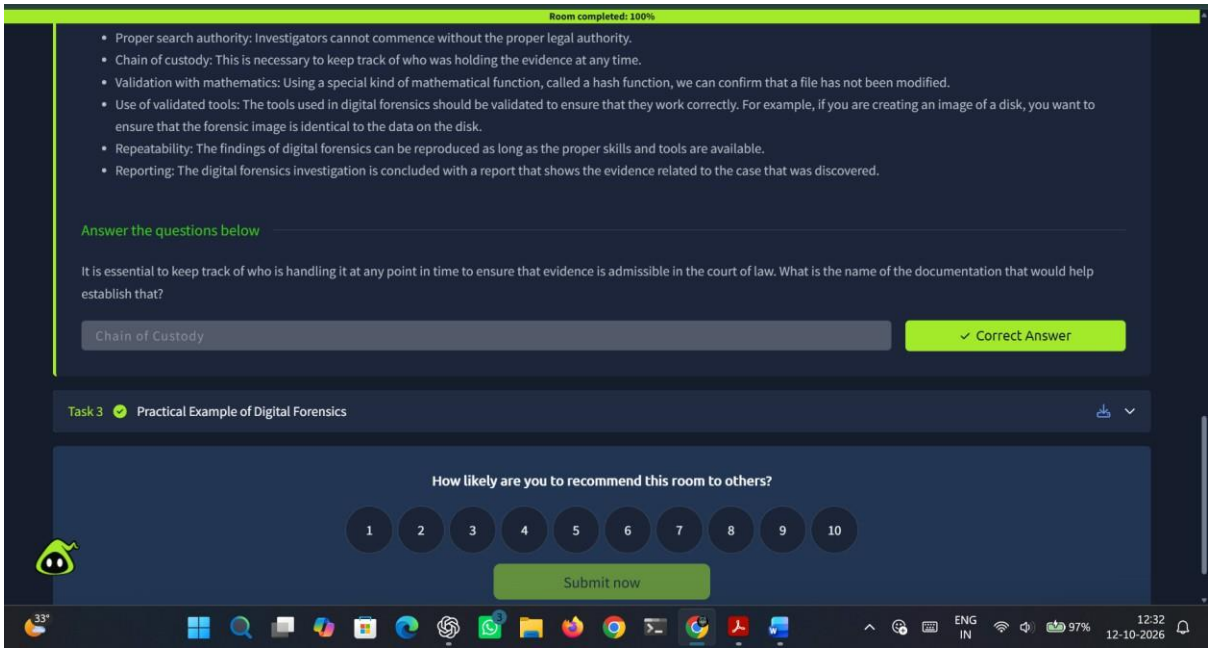


Figure 1.3 Task 2: Digital Forensics Process

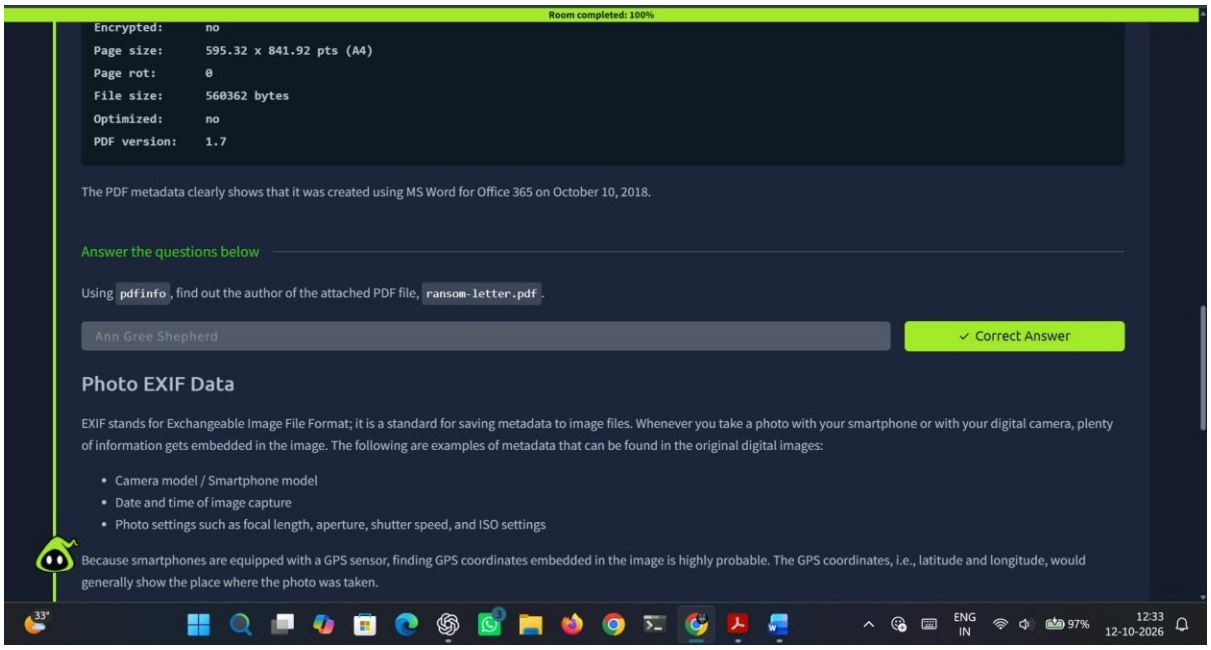


Figure 1.4 Task 3: Practical Example of Digital Forensics

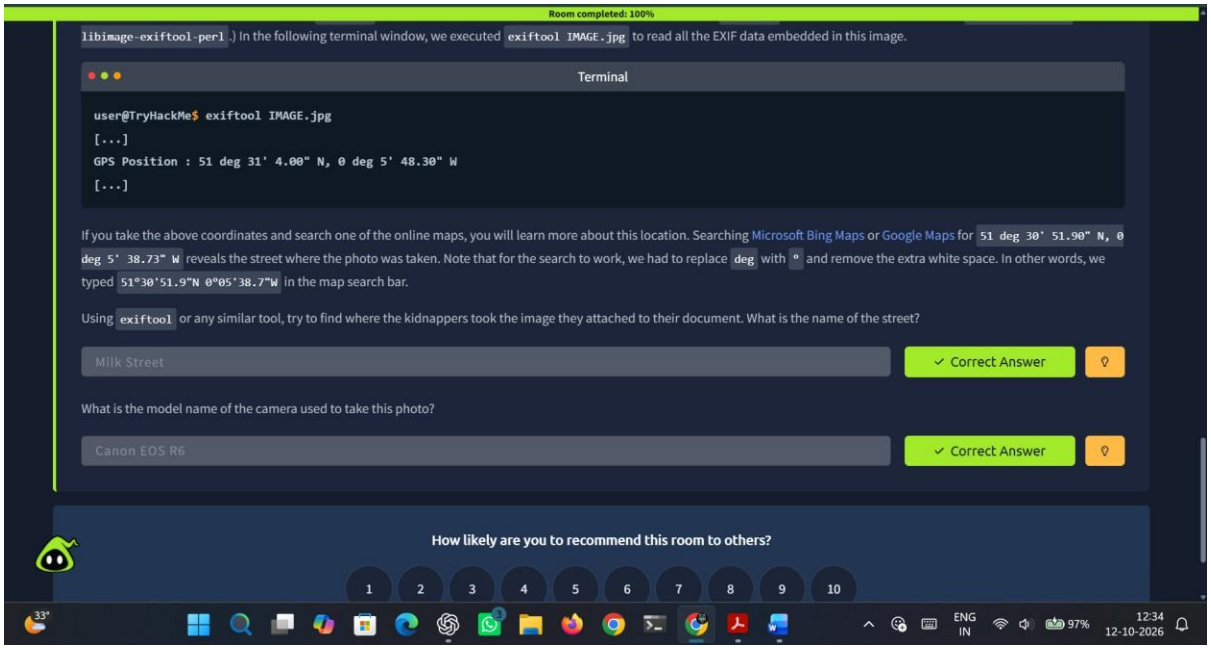


Figure 1.5 Task 3: Practical Example of Digital Forensics

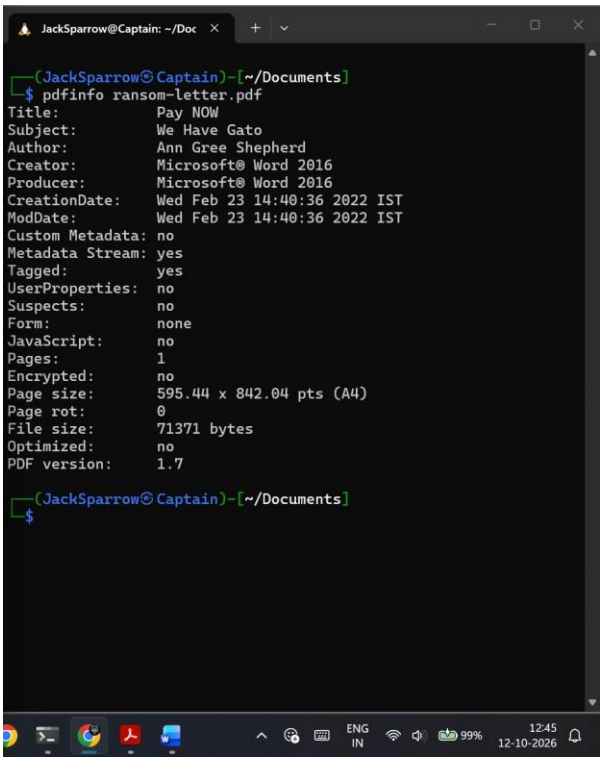


Figure 1.5 View Document Metadata

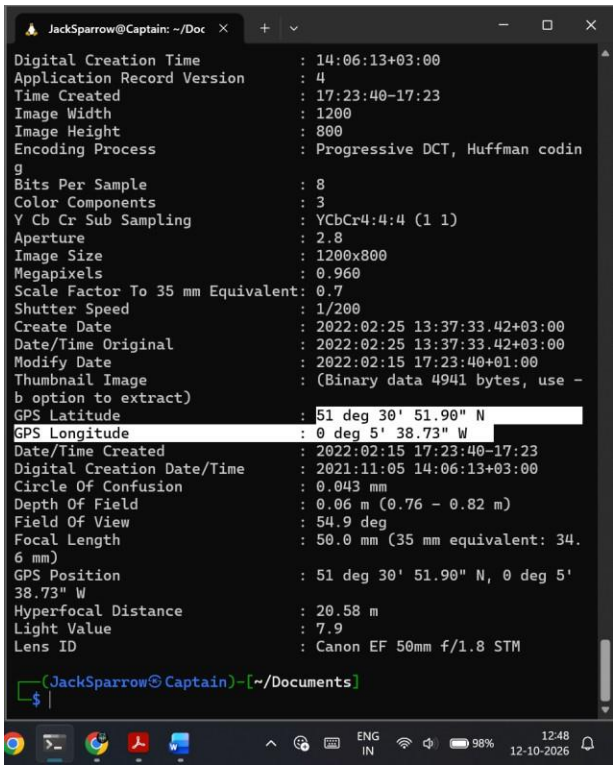


Figure 1.6 View Image metadata (GPS)

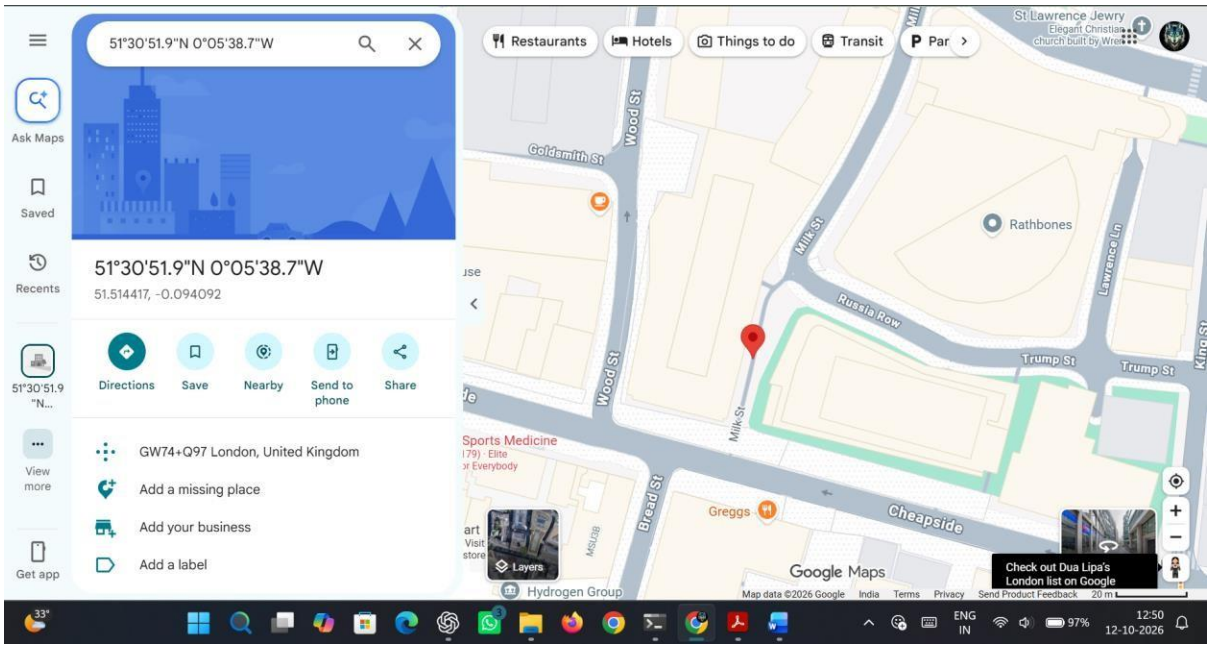


Figure 1.7 Find the street name using Google Maps

RESULT:

Thus the TryHackMe Room 'Intro to Digital Forensics' was completed successfully.

EXPERIMENT NO: 2 **DATE: 16/07/26**

STUDY OF COMPUTER FORENSICS AND DIGITAL FORENSICS TOOLS USED FOR FORENSIC INVESTIGATION

AIM:

To study various digital forensic tools used for collecting, analyzing, preserving, and presenting digital evidence during cyber forensic investigations.

DIGITAL FORENSICS:

Digital forensics is the process of identifying, preserving, collecting, examining, and presenting digital evidence from computers, mobile devices, storage media, and networks while maintaining the integrity of the evidence.

TOOLS STUDIED:

1. Autopsy

Description:

Autopsy is an open-source digital forensic platform that provides a graphical interface for forensic investigations.

Features:

- File recovery
- Deleted file analysis
- Keyword searching
- Timeline analysis
- Hash matching
- Report generation

Applications:

- Hard disk investigations
- Criminal investigations
- Incident response
- Malware analysis

2. Sleuth Kit (TSK)

Description:

The Sleuth Kit is a collection of command-line forensic tools used to analyze disk images and recover files from storage devices.

Features:

- Disk image analysis
- File system analysis
- Deleted file recovery
- Timeline generation
- Metadata examination

Applications:

- Disk forensics
- Data recovery
- File system investigations

3. Wireshark

Description:

Wireshark is a network protocol analyzer used to capture and inspect network packets.

Features:

- Live packet capture
- Protocol analysis
- Packet filtering
- Network troubleshooting
- Traffic statistics

Applications:

- Network forensics
- Intrusion detection
- Malicious traffic analysis

- Security monitoring

4. CAINE

Description:

CAINE (Computer Aided INvestigative Environment) is a Linux-based digital forensic environment.

Features:

- Live forensic analysis
- Disk imaging
- Memory analysis
- Evidence collection
- Data recovery

Applications:

- Live system analysis
- Digital investigations
- Incident response
- Evidence acquisition

5. Volatility

Description:

Volatility is an open-source memory forensics framework used to analyze RAM dumps.

Features:

- Process analysis
- Malware analysis
- DLL analysis
- Network connection analysis
- Registry extraction

Applications:

- Memory forensics

- Malware analysis
- Incident response
- Live memory investigations

6. Plaso (Log2Timeline)

Description:

Plaso (Log2Timeline) is a timeline analysis tool that extracts timestamps from multiple sources to reconstruct system activities.

Features:

- Timeline generation
- Log analysis
- Event reconstruction
- Multiple file format support

Applications:

- Timeline investigations
- Incident response
- Digital forensic analysis

7. ExifTool

Description:

ExifTool is a command-line utility used to read, write, and edit metadata in image, audio, video, and document files.

Features:

- Metadata extraction
- Metadata editing
- GPS information retrieval
- Batch processing

Applications:

- Image forensics

- Document verification
- Photo authentication
- Metadata investigation

RESULT:

Thus, various digital forensic tools, including Autopsy, Sleuth Kit, Wireshark, CAINE, Volatility, Plaso, and ExifTool, were studied along with their features and applications in digital forensic investigations.

EXPERIMENT NO: 3 **DATE: 23/07/2026**

WINDOWS FORENSICS 1

AIM:

To study about Windows Registry forensics.

PROCEDURE:

1. Open Registry Explorer and load SYSTEM, SOFTWARE, SAM and NTUSER.DAT hives.
2. Check computer name in Control\ComputerName and user accounts from SAM ...\Users.
3. Find recent files in RecentDocs and typed commands in RunMRU.
4. Verify program execution via UserAssist and BAM keys.
5. Identify USB drives in USBSTOR and volume labels in Portable Devices.
6. Fill in all the answers and complete the room.

TASKS:

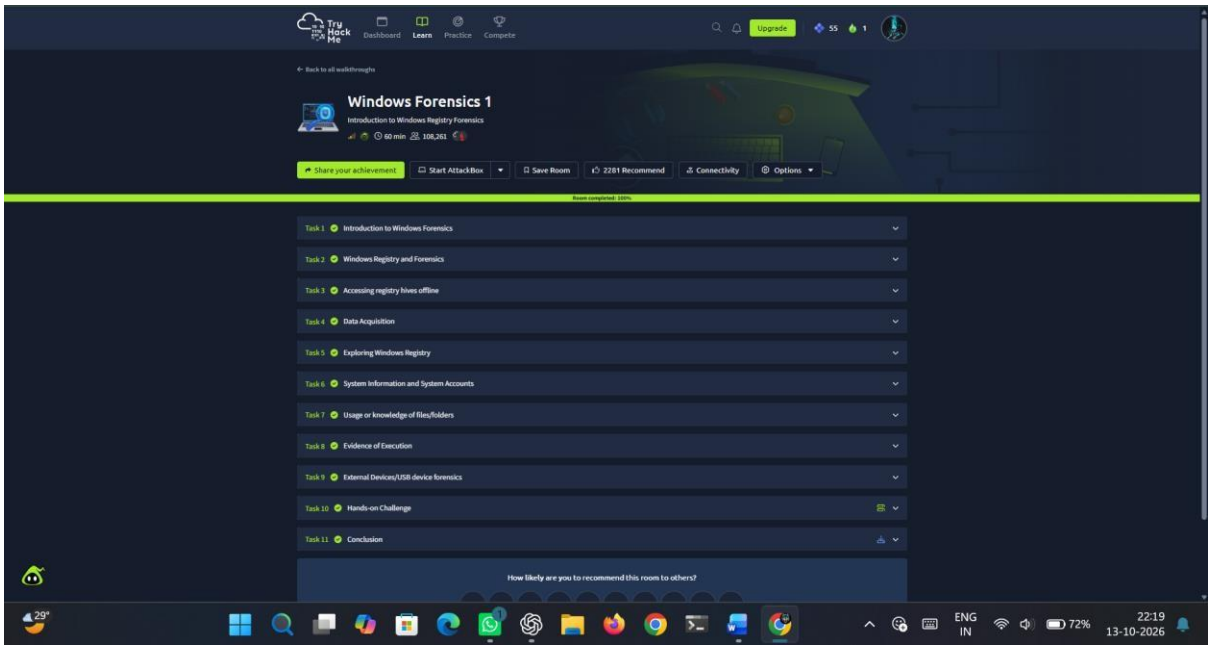


Figure 3.1 Windows Forensics 1-Tryhackme room

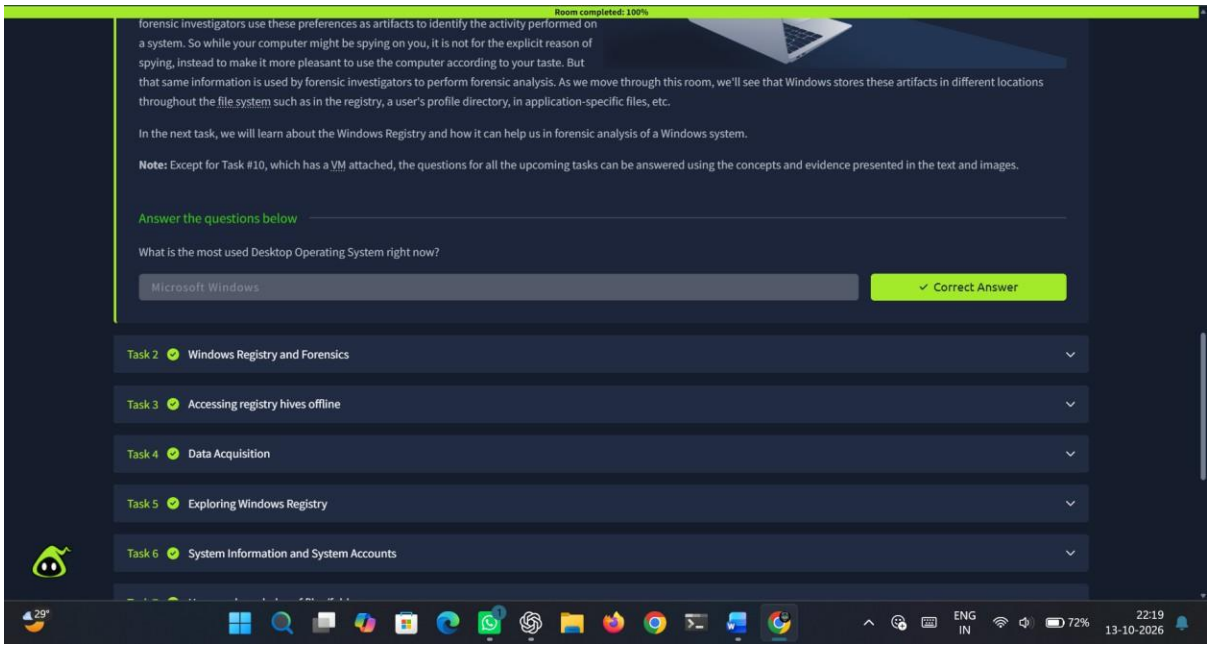


Figure 3.2 Task 1

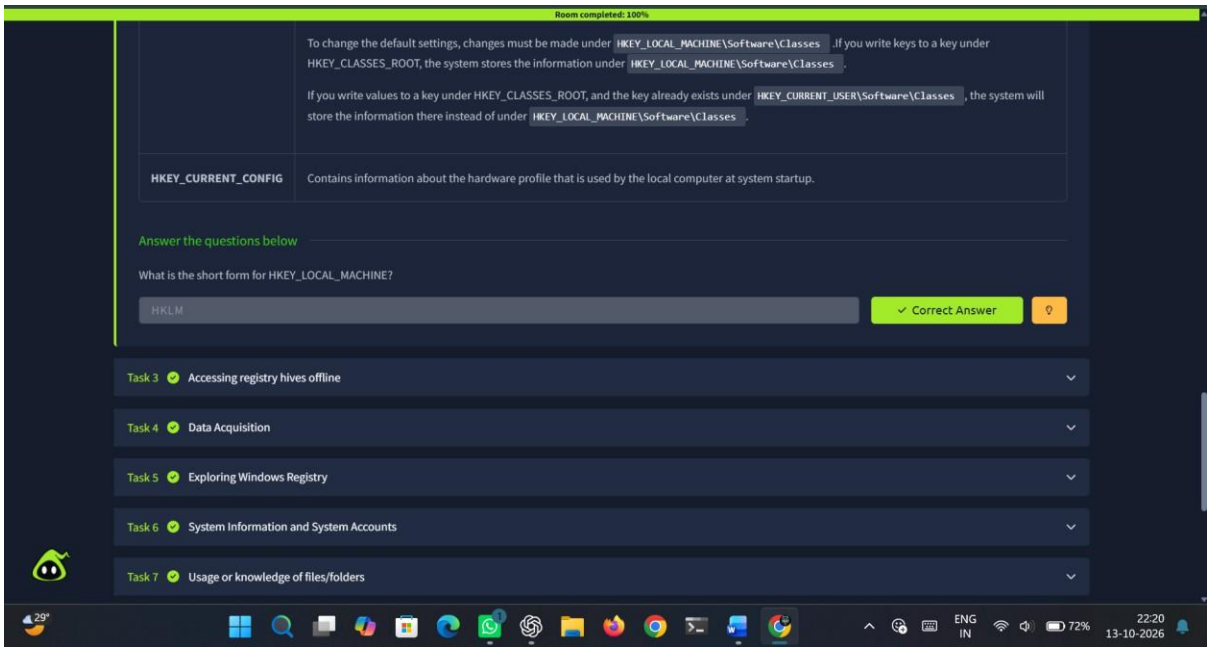


Figure 3.3 Task 2

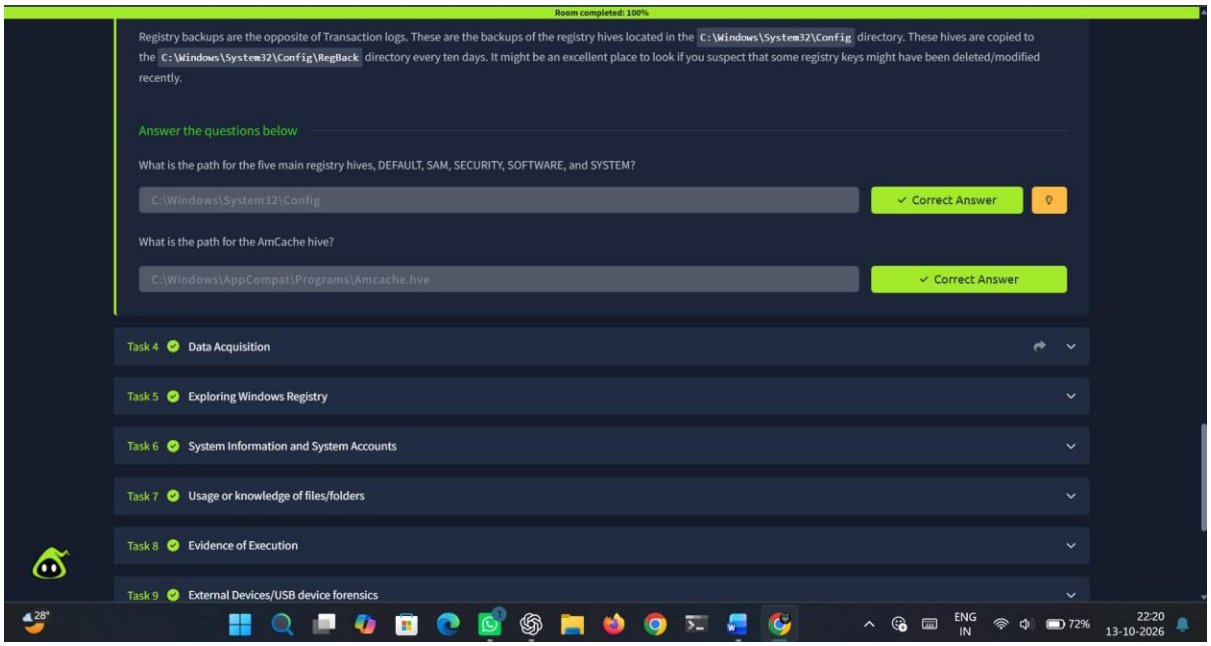


Figure 3.4 Task 3

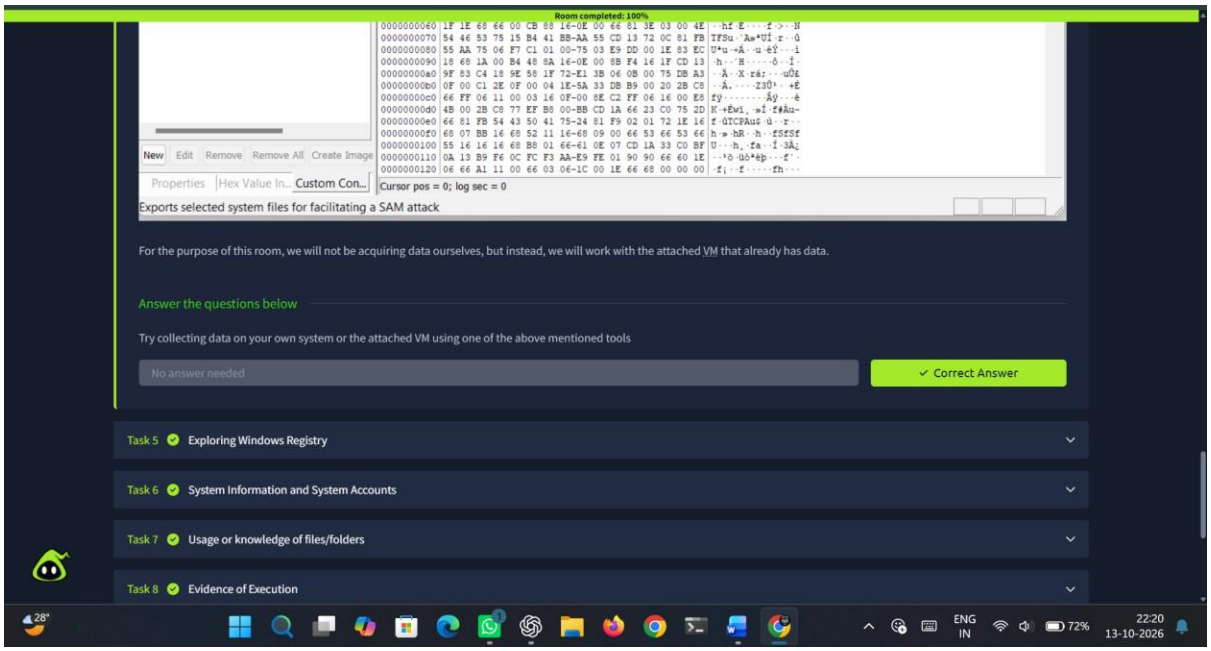


Figure 3.5 Task 4

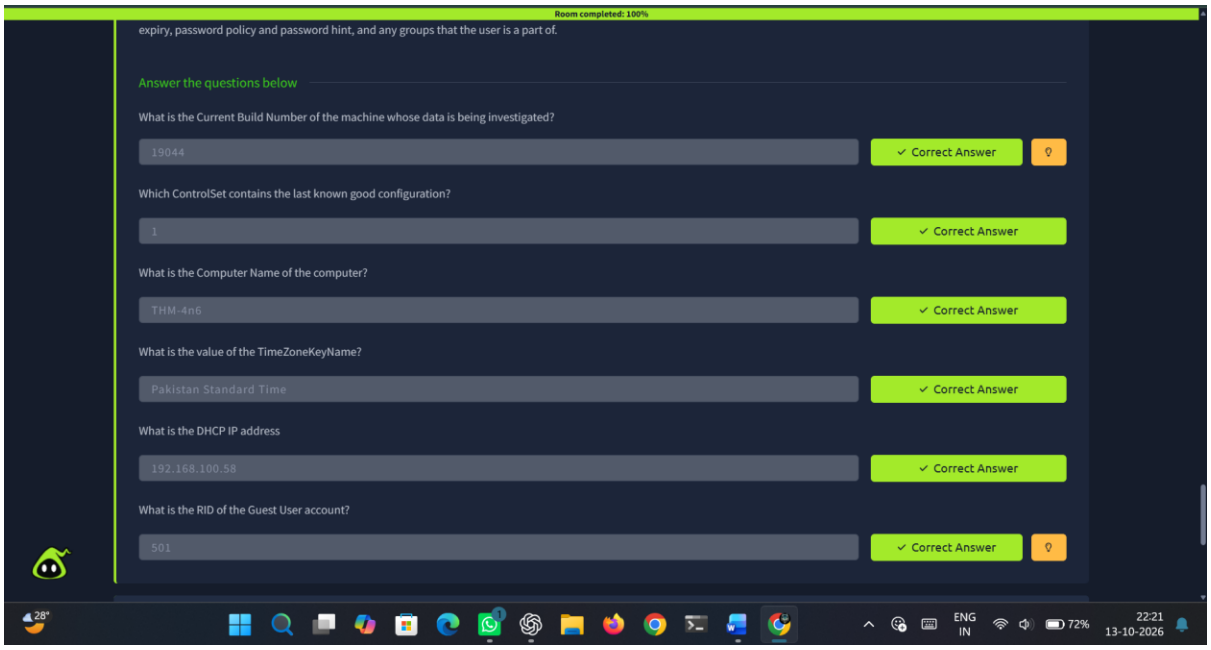


Figure 3.6 Task 6

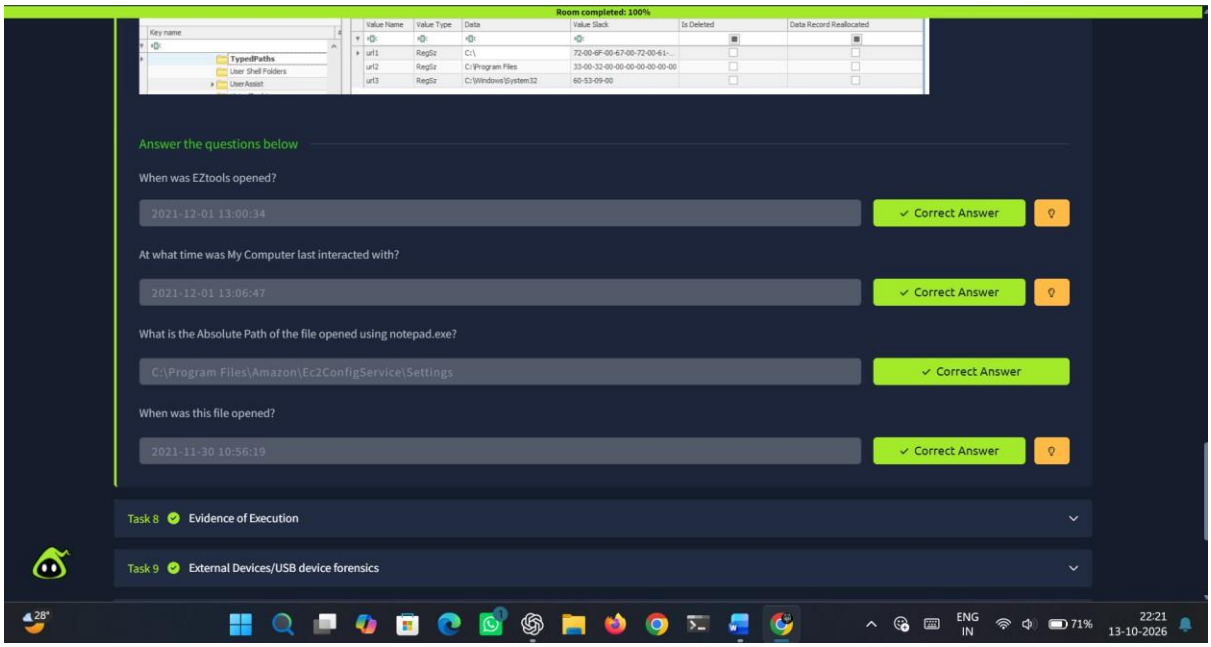


Figure 3.7 Task 7

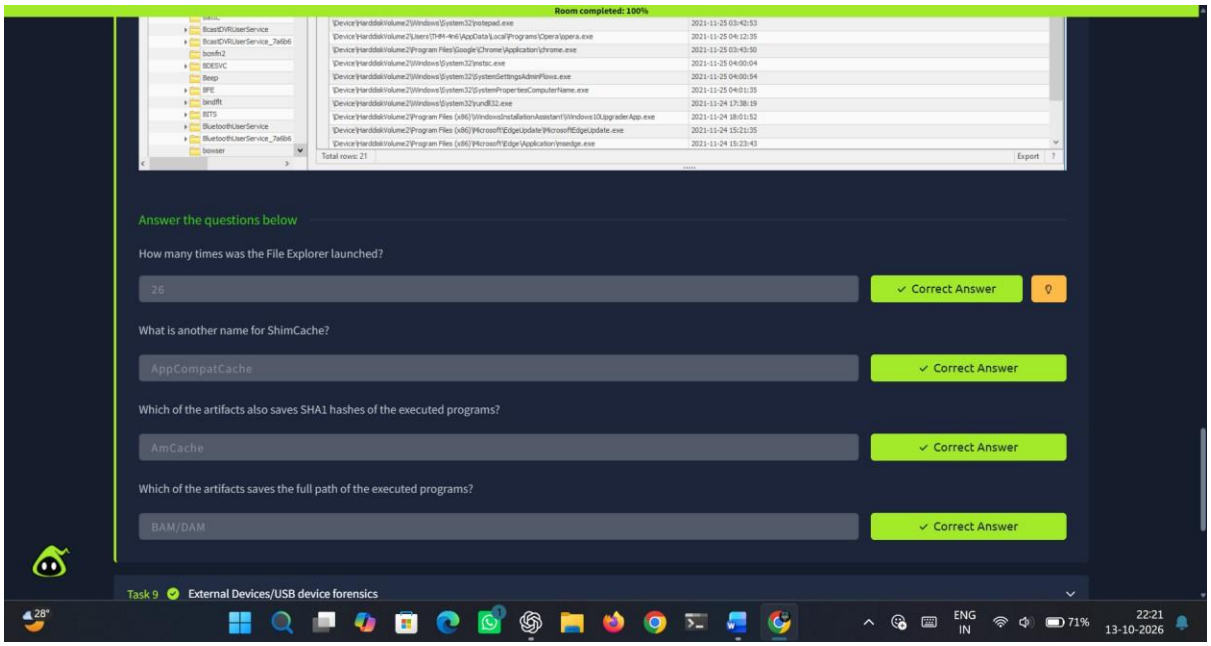


Figure 3.8 Task 8

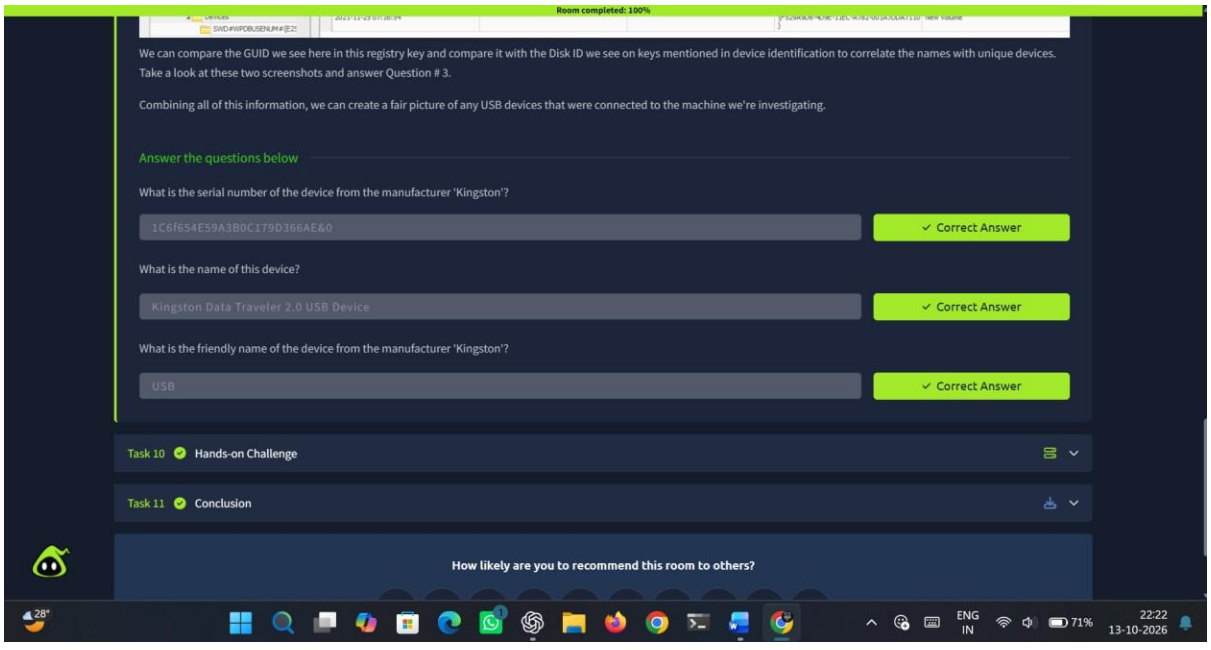


Figure 3.9 Task 9

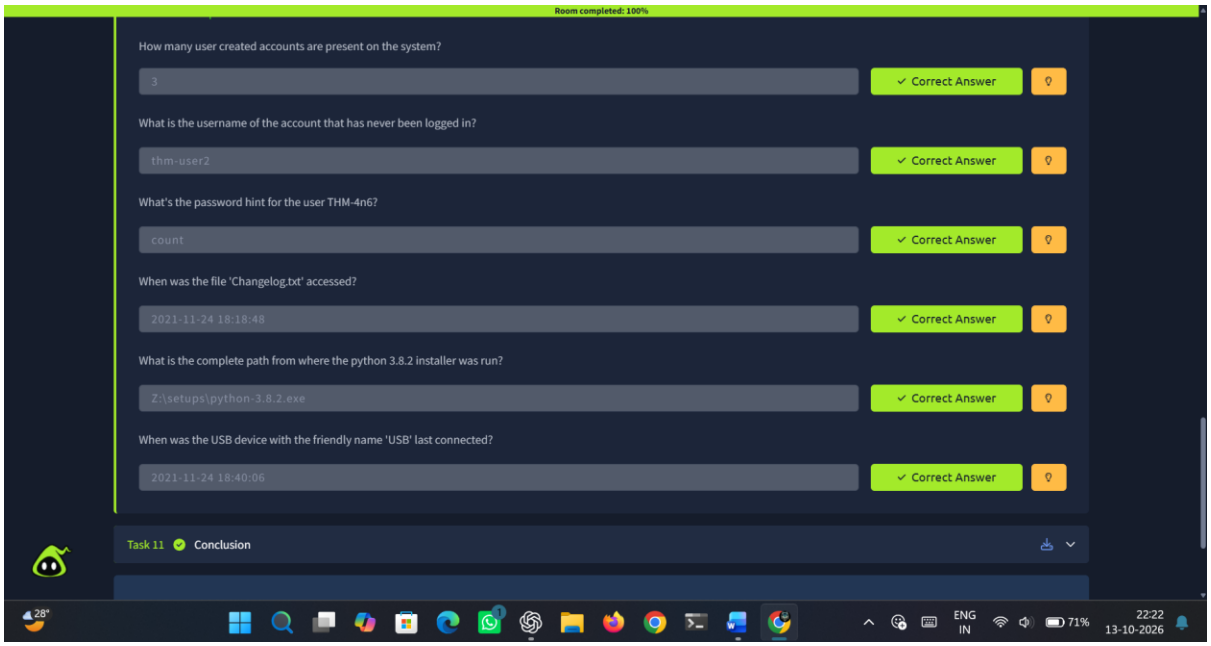


Figure 3.10 Task 10

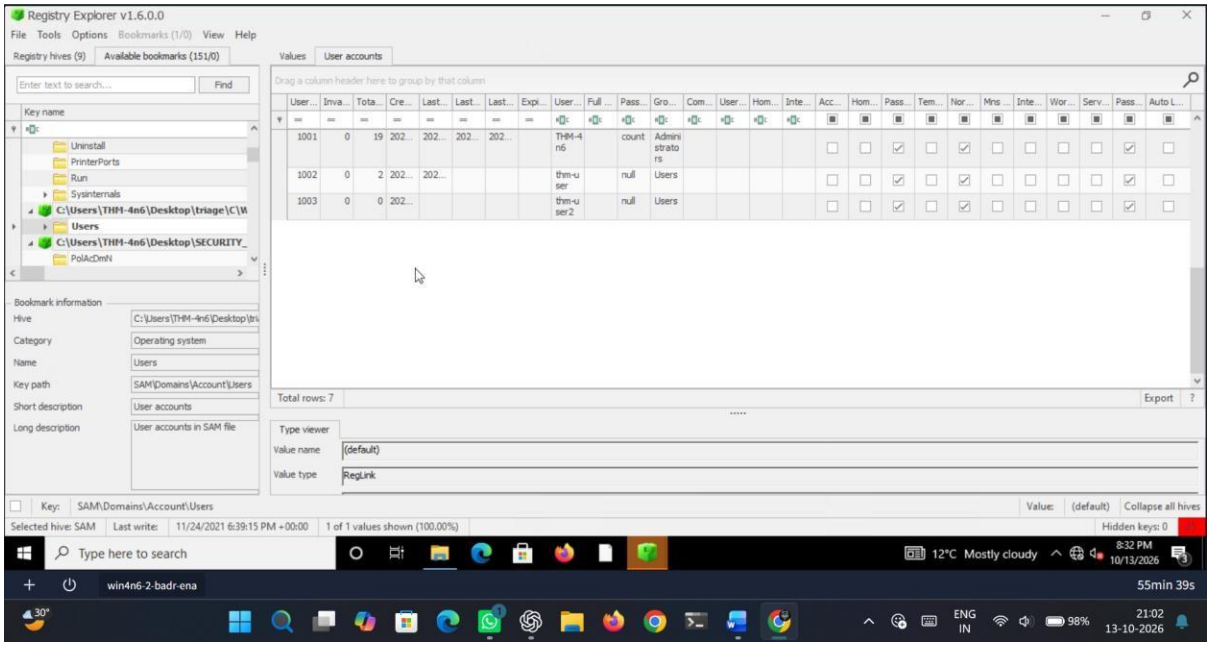


Figure 3.11 User accounts

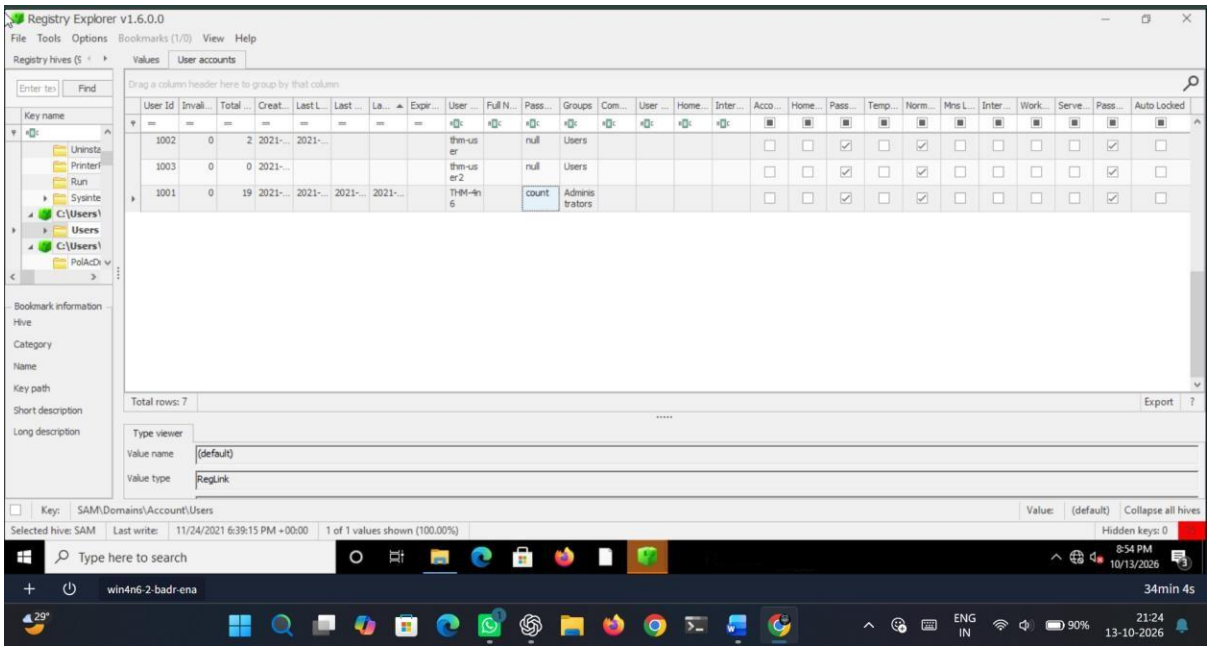


Figure 3.12 User account password

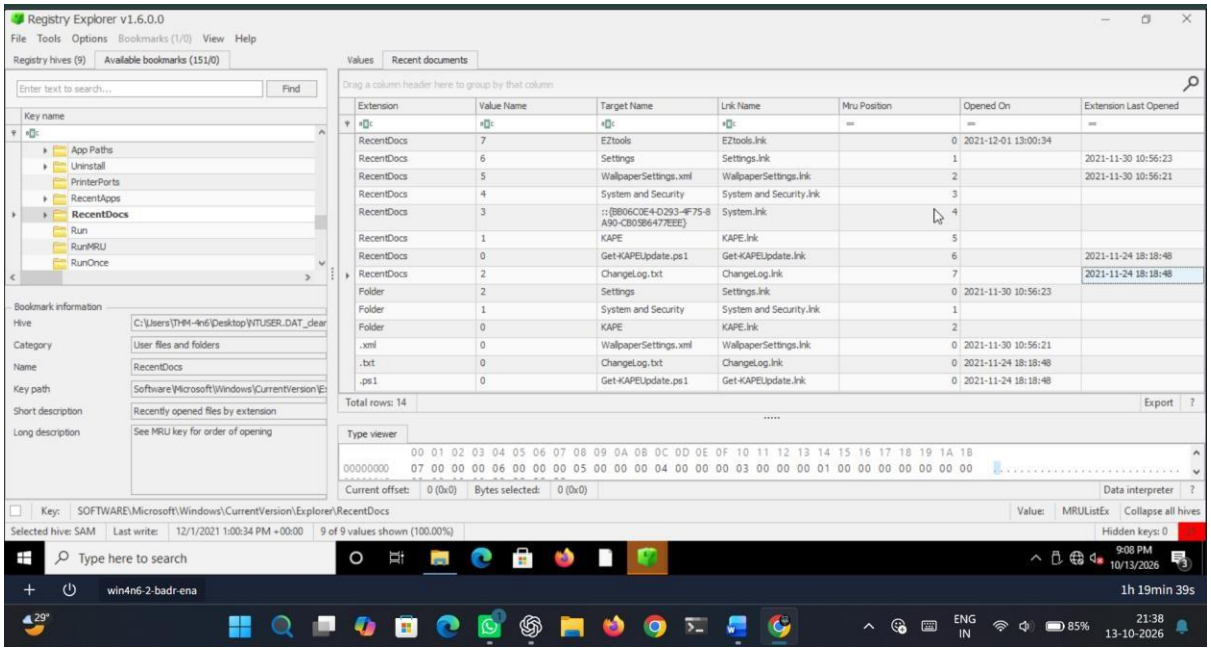


Figure 3.13 Last opened timestamp for 'Changelog.txt'

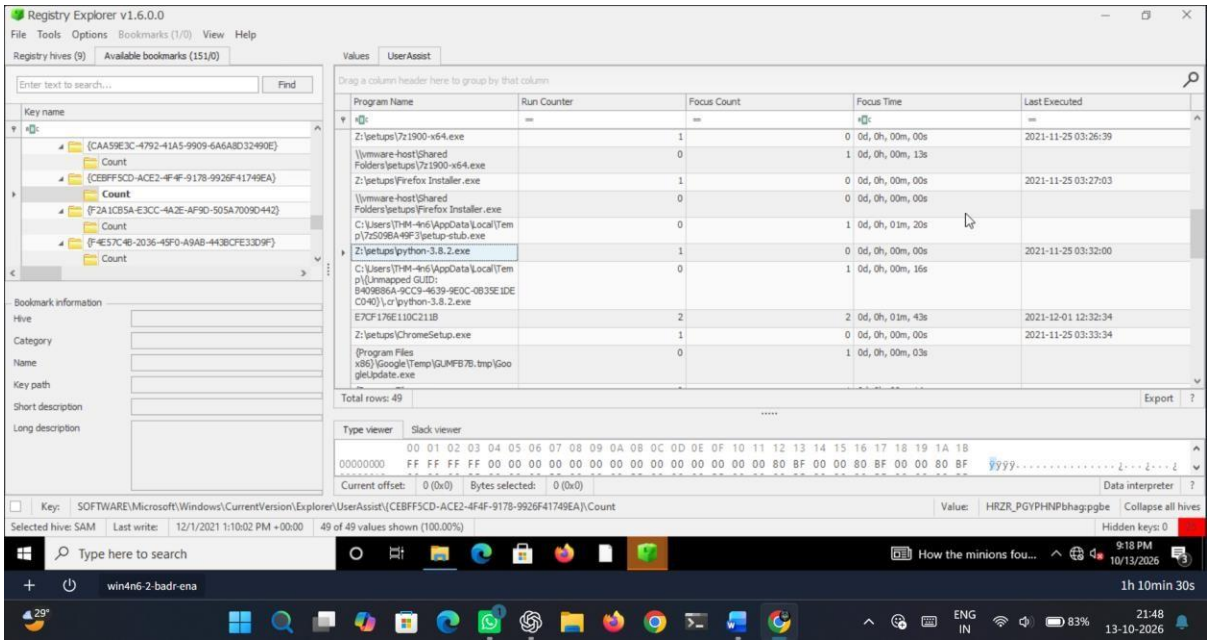


Figure 3.14 Complete path of python installer

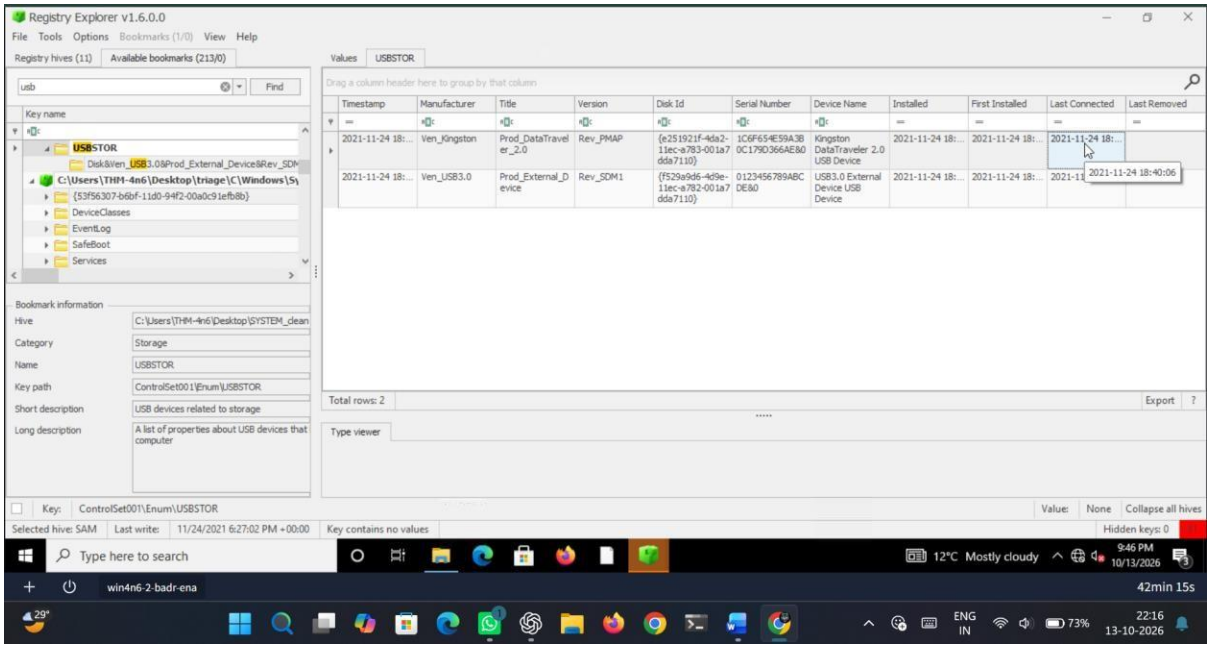


Figure 3.15 USB timestamp

RESULT:

Thus the TryHackMe Room 'Windows Forensics 1' was completed.

EXPERIMENT NO: 4 **DATE: 30/7/2026**

LIVE FORENSIC INVESTIGATION USING AUTOPSY

AIM

To perform a basic forensic investigation of a Windows system using **Autopsy** and identify files, user activity, browser history, deleted files and other digital evidence.

SOFTWARE REQUIRED

- Windows 10/11
- Autopsy
- One authorized laboratory/test computer
- USB drive, if required

PROCEDURE

Step 1 – Open Autopsy

1. Start **Autopsy**.
2. Select **Create New Case**.
3. Enter:
    - Case Name: Live_Forensics
    - Case Number: LF001
    - Examiner Name: Your Name
4. Click **Finish**.

Step 2 – Add Evidence

1. Click **Add Data Source**.
2. Select **Local Disk**.
3. Select the authorized test computer's **C: drive**.
4. Click **Next**.

Important: Perform this only on the laboratory/test computer authorized by your instructor.

Step 3 – Select Ingest Modules

Select the following if available:

- Recent Activity
- File Type Identification

- Keyword Search
- Hash Lookup

Click **Next / Finish**.

Wait for the ingest process to complete.

Step 4 – Examine User Files

From the left panel, navigate to

Data Sources → C: → Users

Open the test user's folders:

- Desktop
- Documents
- Downloads
- Pictures

Identify any interesting files.

Record:

File Name	Location	File Type
325224910	C:/Users/ponsu/Documents/NOTES/ds/325224910.pdf	.pdf

Step 5 – Examine Browser Activity

Go to

Results → Recent Activity

Look for

- Web history
- URLs
- Downloads
- Browser information

Record any two relevant entries.

Step 6 – Search for a Keyword

- Use **Keyword Search**.
- Search for a simple laboratory keyword such as:

  ProjectX or Evidence

- Record the result.

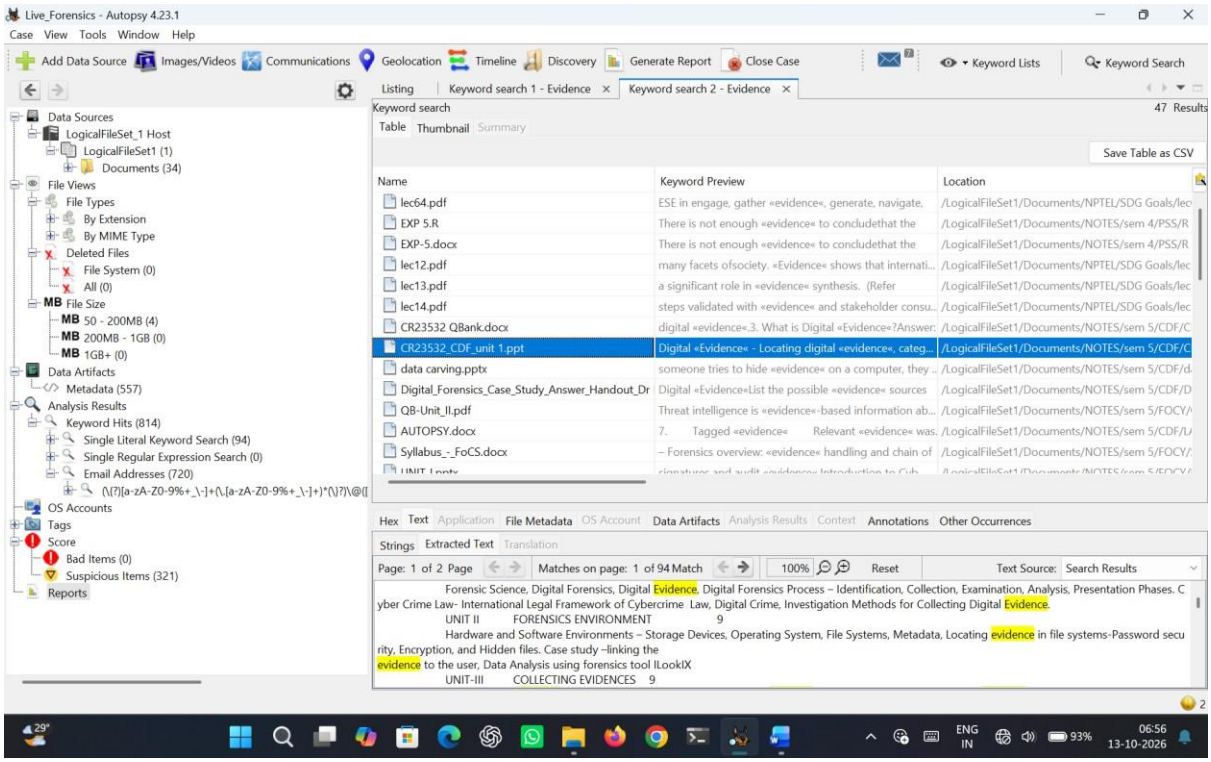


Figure 4.1 Result for keyword search 'evidence'

Keyword	File	Location
evidence	CR23532_CDF_unit 1.ppt	C:/Users/ponsu/Documents/NOTES/sem 5/ CR23532_CDF_unit 1.ppt

Step 7 – Examine Deleted Files

Look under the available **Deleted Files** results.

Find one deleted file, if available.

Record:

- File name
- File type
- Location
- Size
- Deleted status

Step 8 – Examine File Details

Select one important file.

Check the **Content Viewer / File Details**.

Record:

- File name
- Path
- Size
- Created time
- Modified time
- Accessed time

Step 9 – Tag Important Evidence

Right-click the important file.

Select:

Tag → Interesting Item / Notable Item

Add a short comment such as:

"File identified as relevant evidence during the investigation."

Step 10 – Generate Report

Go to **Tools → Generate Report**

Select:

HTML Report

Generate the report.

Open the generated report and verify the evidence.

OBSERVATION:

S.No.	Evidence Examined	Finding
1.	User files	User folders were successfully examined and relevant user files were identified.
2.	Browser history	No browser history found.
3.	Downloaded files	No downloaded files were found.
4.	Keyword search	The keyword search located files containing the specific keyword, confirming the presence of matching data.
5.	Deleted files	No deleted files found.
6.	File timestamps	Created, modified, accessed timestamps of selected files were examined.
7.	Tagged evidence	Relevant evidence was tagged and documented.

SCREENSHOTS:

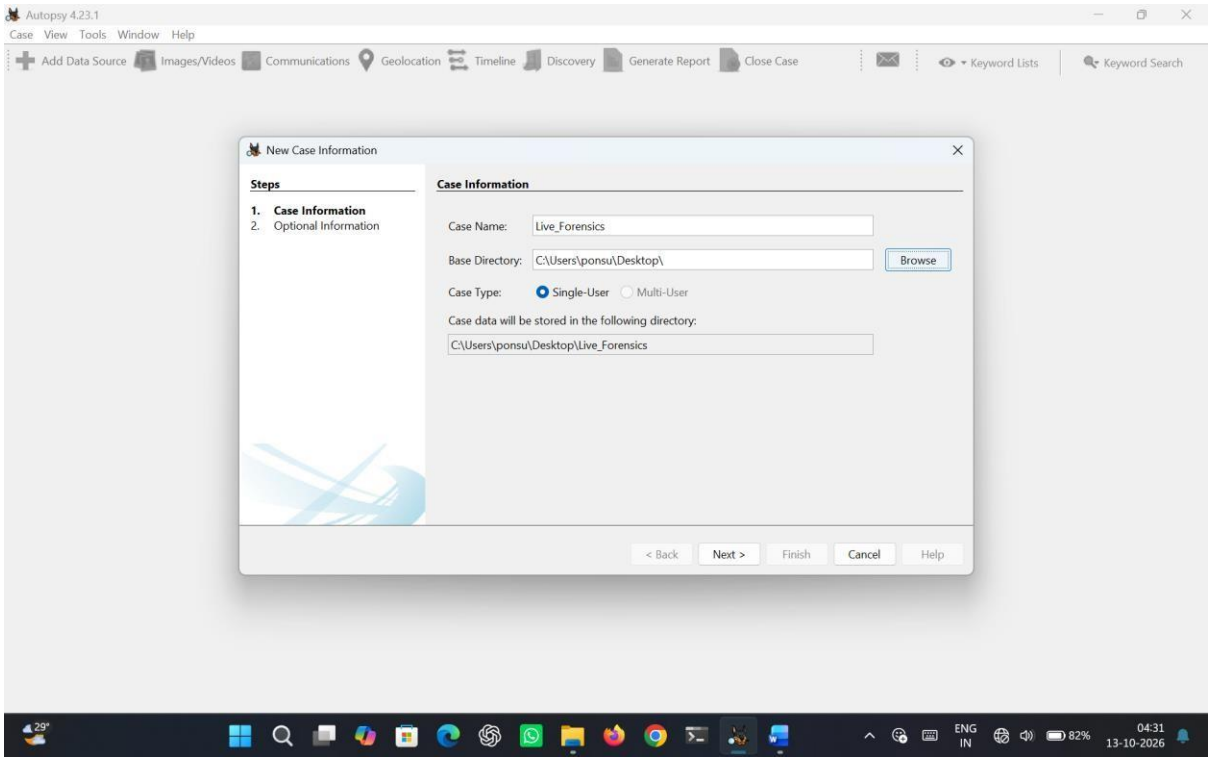


Figure 4.2 New case

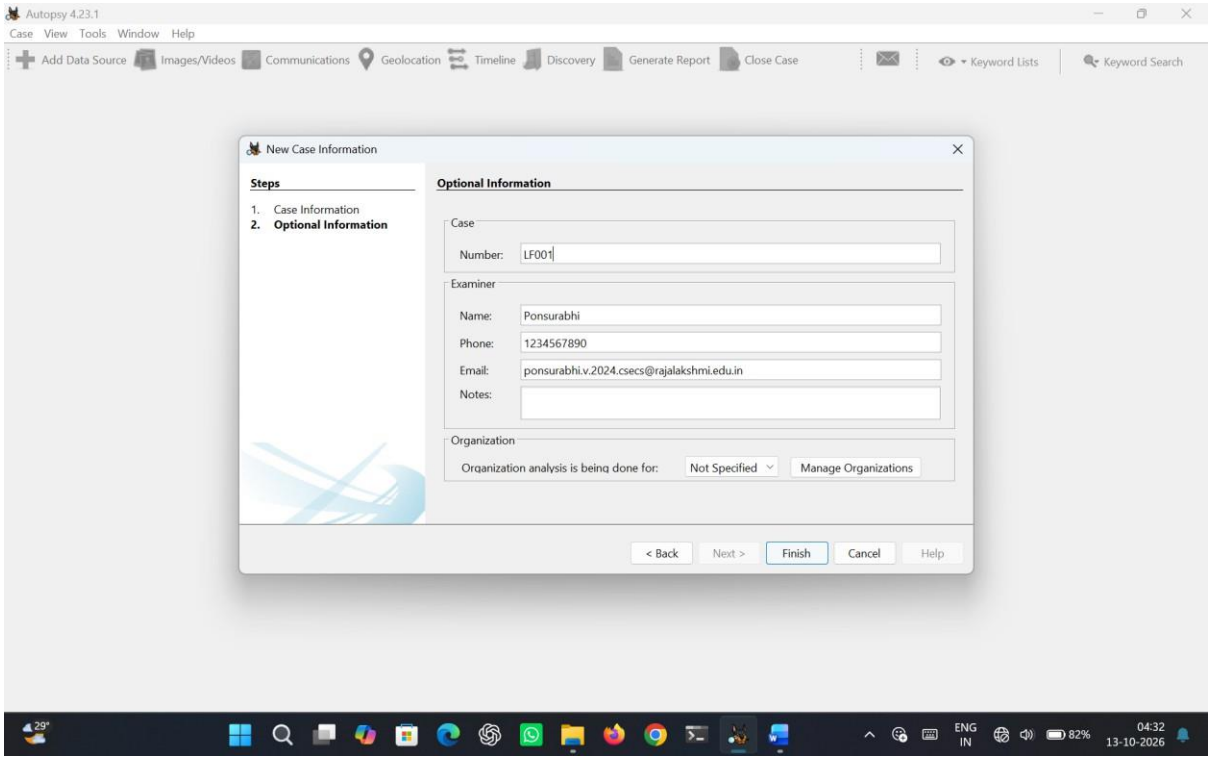


Figure 4.3 Case details

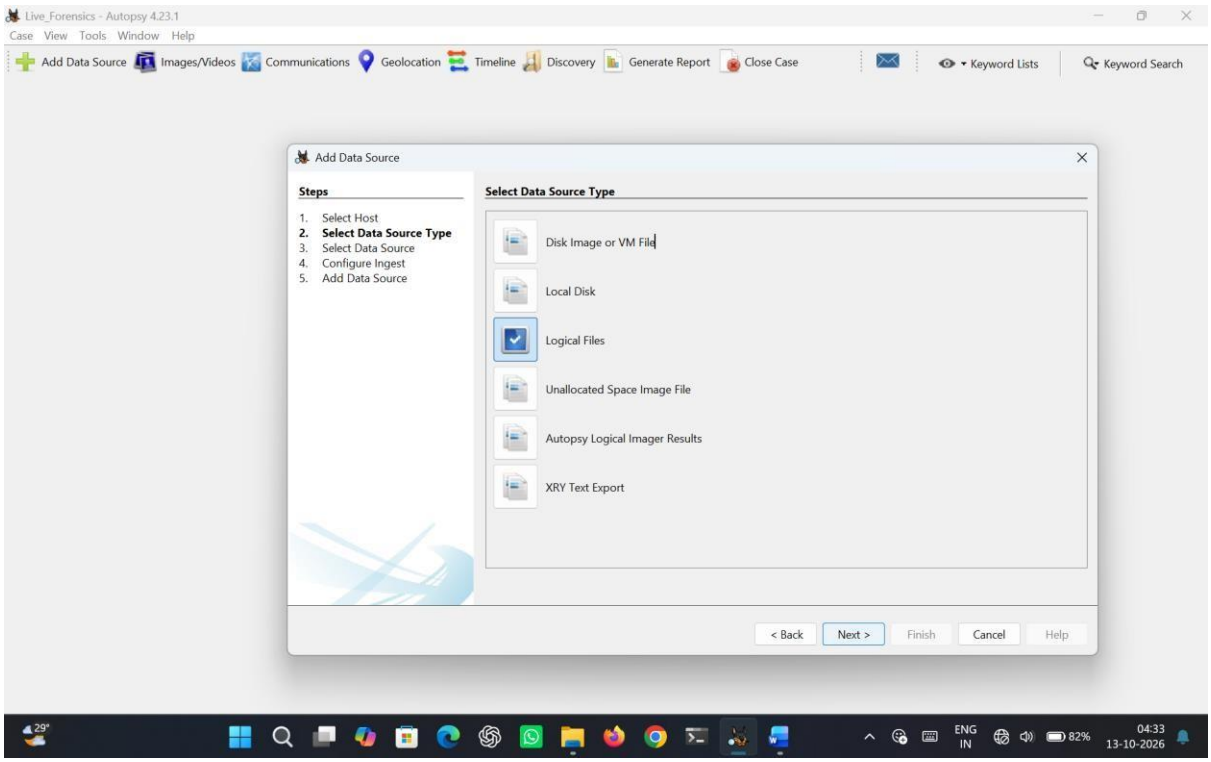


Figure 4.4 – Data source type

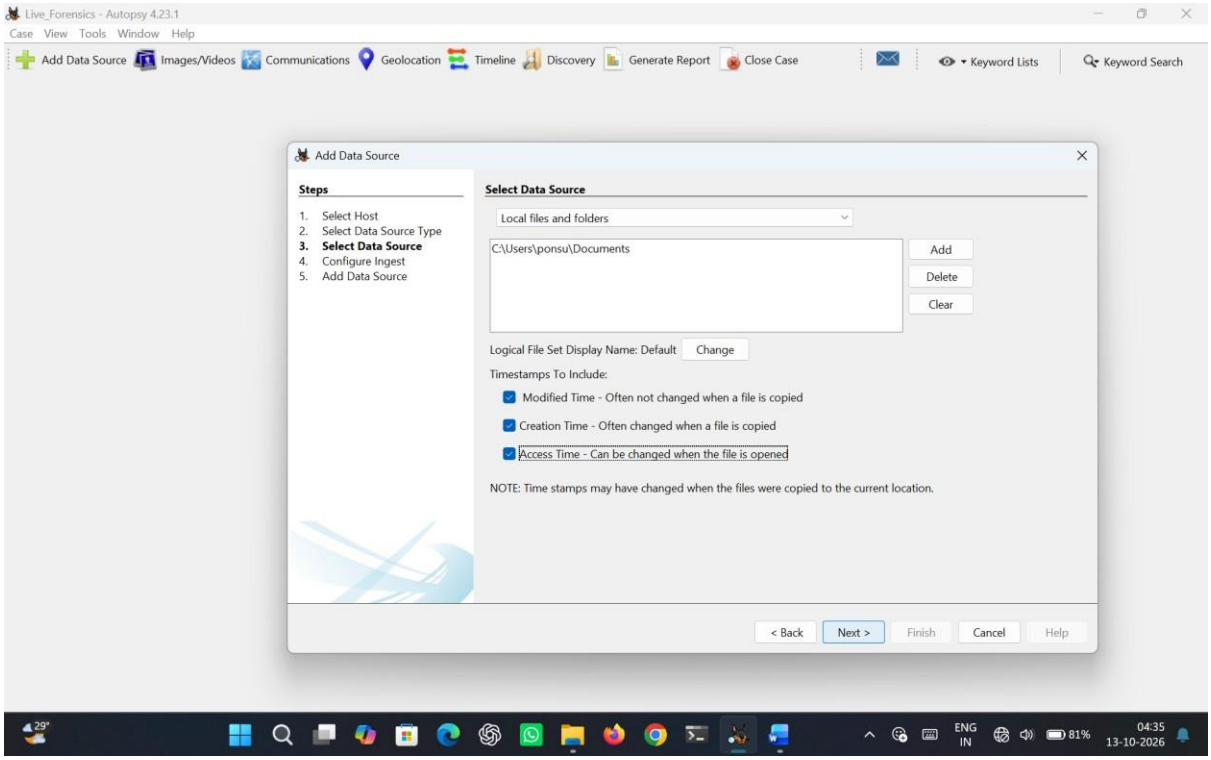


Figure 4.5 Select Data source

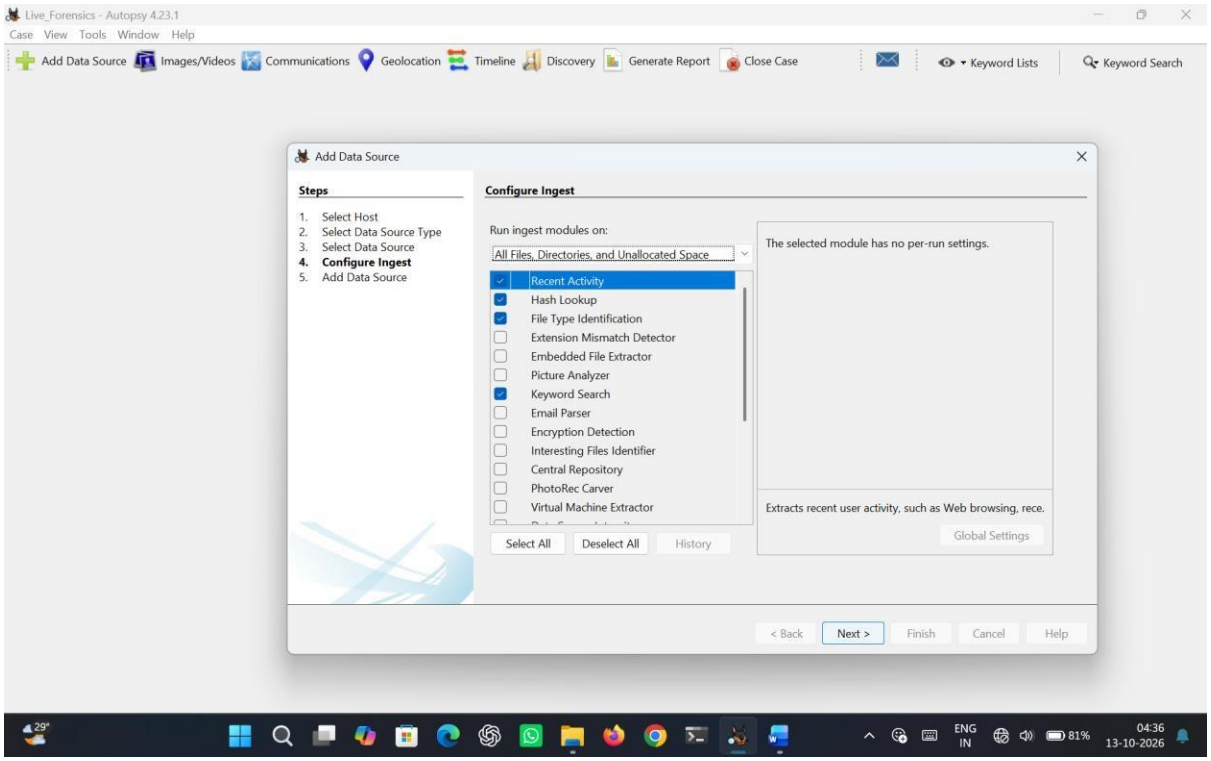


Figure 4.6 Configure Ingest

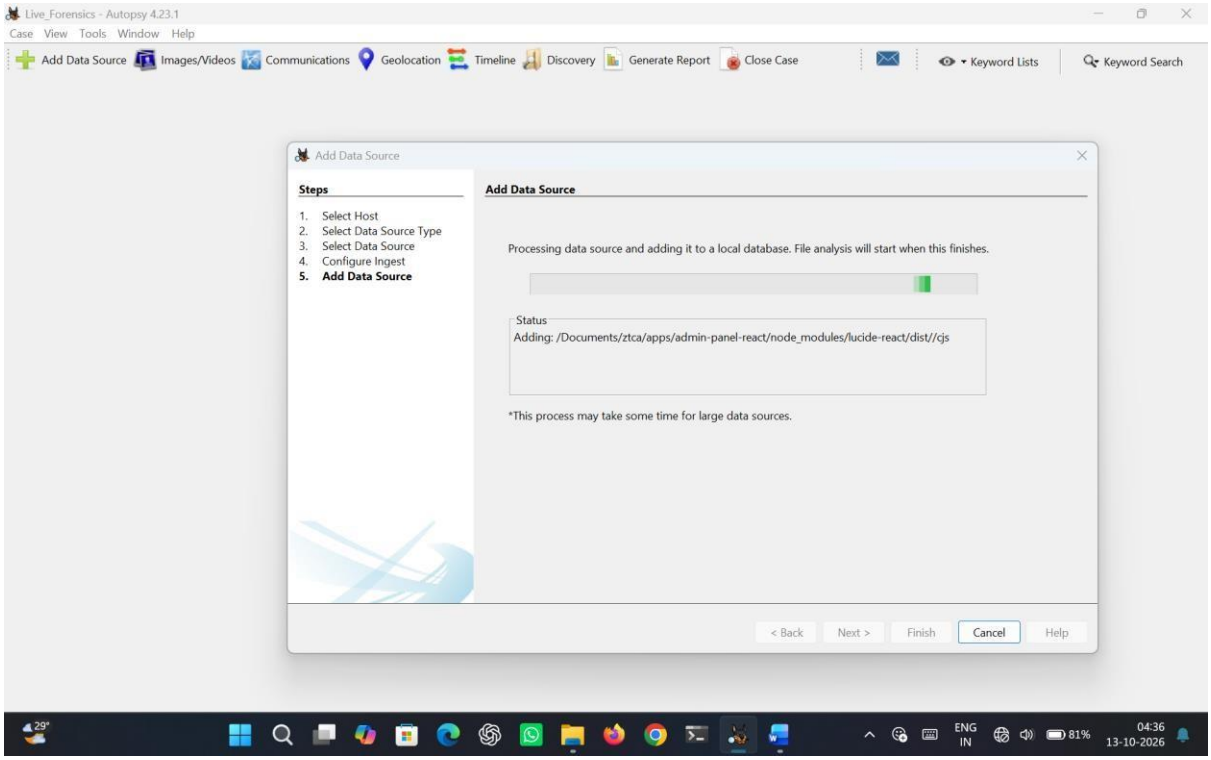


Figure 4.7 Add data source

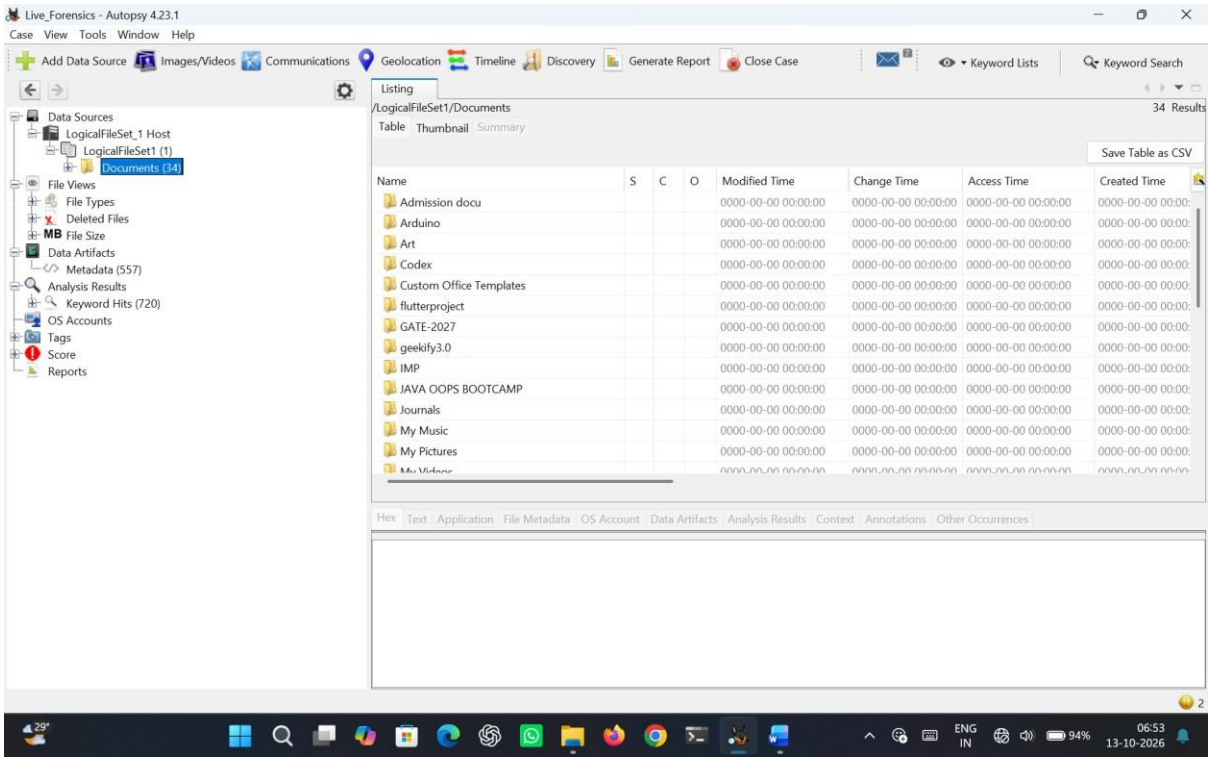


Figure 4.8 Results

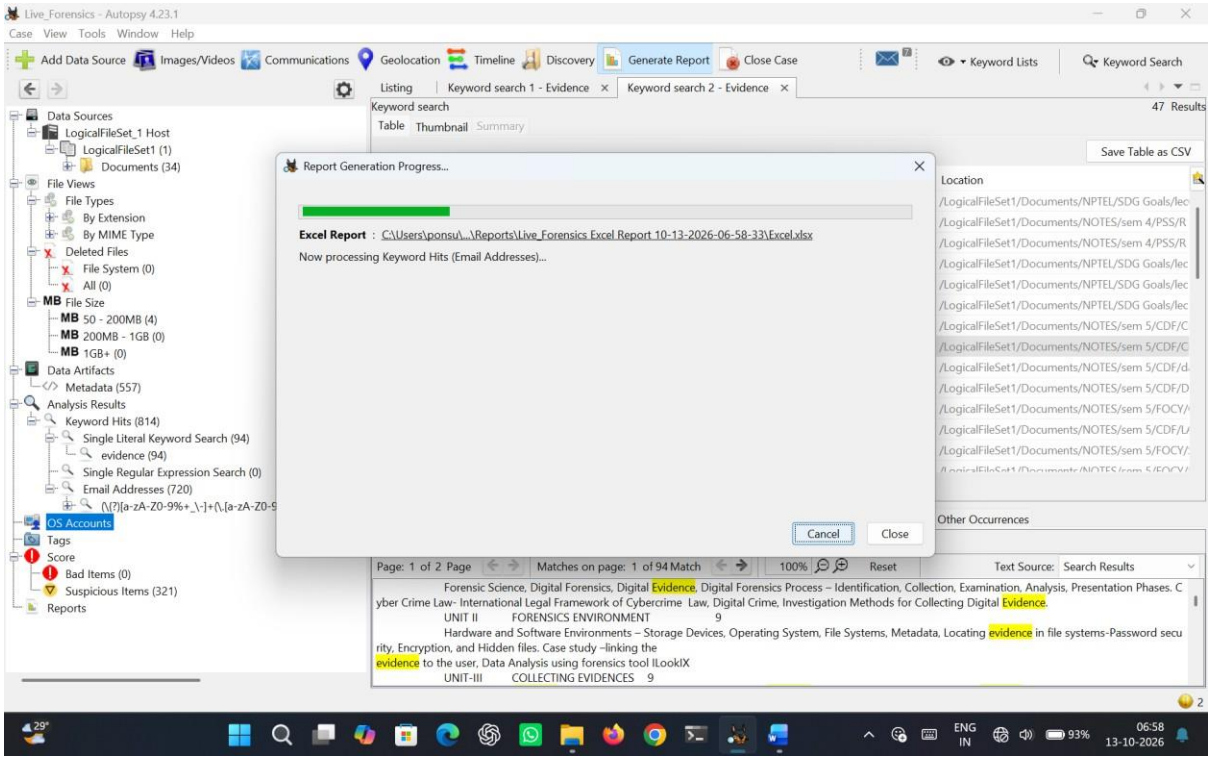


Figure 4.9 Generate Report

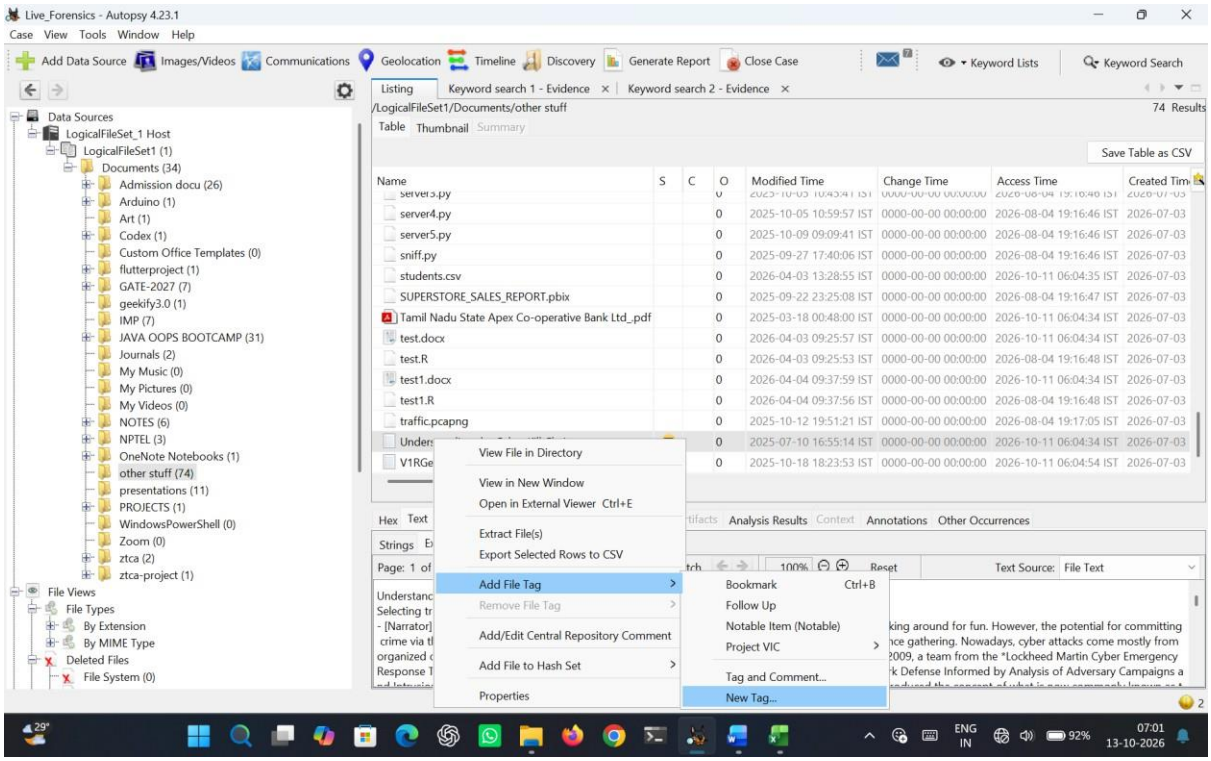


Figure 4.10 Tag Interesting Item

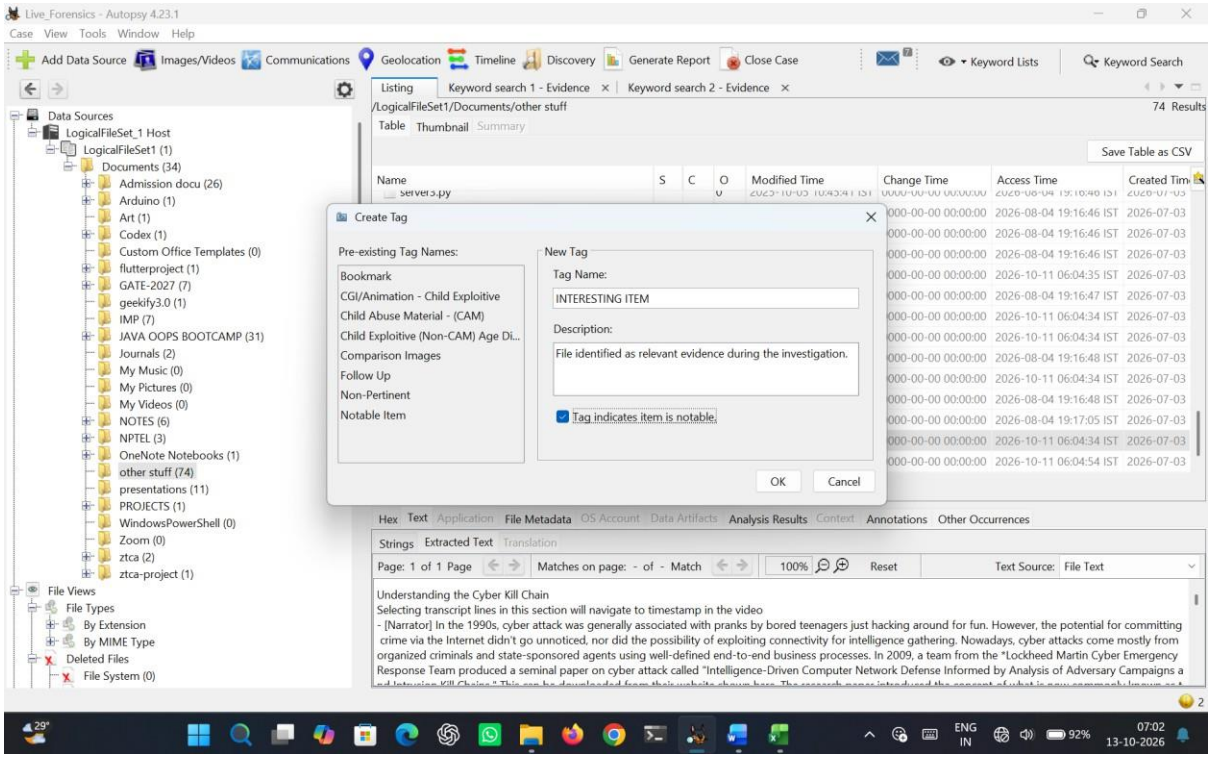


Figure 4.11 Create Tag

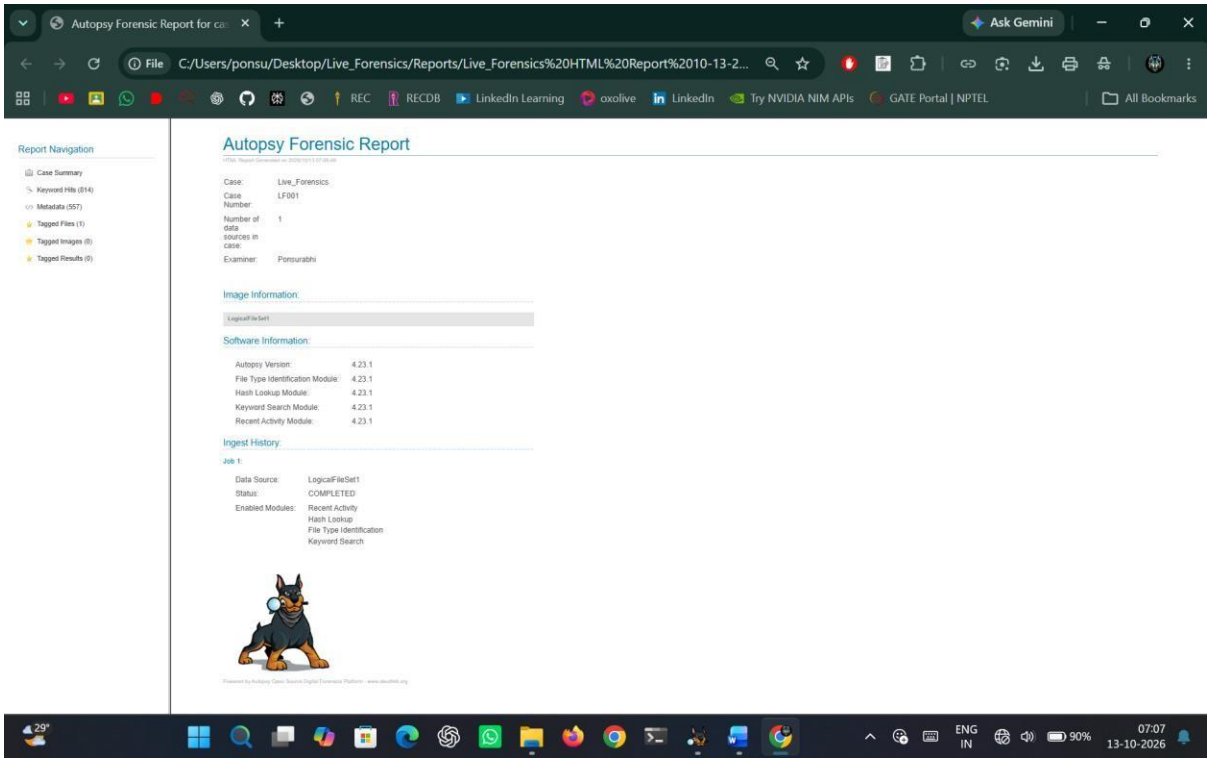


Figure 4.12 Autopsy Forensic Report

RESULT

The Windows test system was successfully investigated using **Autopsy**. Relevant digital evidence including user files, browser activity, deleted files and file metadata was identified, examined and documented. A forensic HTML report was successfully generated.

EXPERIMENT NO: 5 **DATE: 06/08/2026**

FTK IMAGER - DISK IMAGING AND EVIDENCE ACQUISITION

AIM

To acquire a forensic image of a storage device using FTK Imager while preserving the integrity of digital evidence.

OBJECTIVES

- Understand forensic disk acquisition.
- Create a forensic image using FTK Imager.
- Generate hash values (MD5/SHA1).
- Verify image integrity.
- Document the acquisition process.

TOOLS & REQUIREMENTS

- FTK Imager
- Windows 10/11
- USB Drive (4 GB or above) or Sample Evidence Image
- External Storage Device (for saving the image)
- Write Blocker (Recommended)

THEORY

FTK Imager is a digital forensic acquisition tool used to create exact bit-by-bit copies of storage media. It captures allocated, deleted, and unallocated data without modifying the original evidence. Hash values are generated during acquisition to ensure evidence integrity.

Source Drive (Evidence) → FTK Imager (Forensic Acquisition) → Forensic Image (E01 Format)

E01 IMAGE FORMAT

- Expert Witness Compression Format.
- Stores data, metadata, and hash values.

- Widely accepted in digital forensics.
- Supports segmenting large images.

BEST PRACTICES

- Always use a write blocker.
- Never analyze the original evidence.
- Store images in secure media.
- Maintain chain of custody.
- Verify hash values after acquisition.

PROCEDURE

Step 1: Launch FTK Imager.

Step 2 : Select File → Create Disk Image.

Step 3 : Choose the source type:

- Physical Drive
- Logical Drive
- Image File
- Contents of a Folder

Click Next.

Step 4: Select the required source drive.

Step 5: Click Add to specify the destination.

Step 6: Choose E01 (Expert Witness Format).

Step 7: Enter case details:

1. Case Number: CASE-001
2. Evidence Number: EVD-001
3. Examiner: Student Name
4. Description: USB Drive Image

Click Next.

Step 8: Specify the destination folder and image filename.

Step 9: Enable verification and hash calculation:

Verify images after acquisition

- Calculate MD5
- Calculate SHA1

Click Start.

Step 10: Click Start to begin acquisition.

Step 11: Wait until imaging completes successfully.

Step 12: Note the generated MD5 and SHA1 hash values.

OBSERVATIONS:

1. Evidence Source: USB Drive Image (hp x796w USB Device)
2. Image Format: E01 (EnCase Evidence File)
3. Image Size: 14,800 MB
4. MD5 Hash: b30428587f6ebcc57d5693e4c0642594
5. SHA1 Hash: 5d98684ddc5a5f8c471df0cc7e1d07c586e154b0
6. Acquisition Time: 13-10-2026, 04:10:06 AM – 04:14:12 AM
7. Verification Status: Verified (MD5 and SHA1 checksums matched).

SCREENSHOTS:

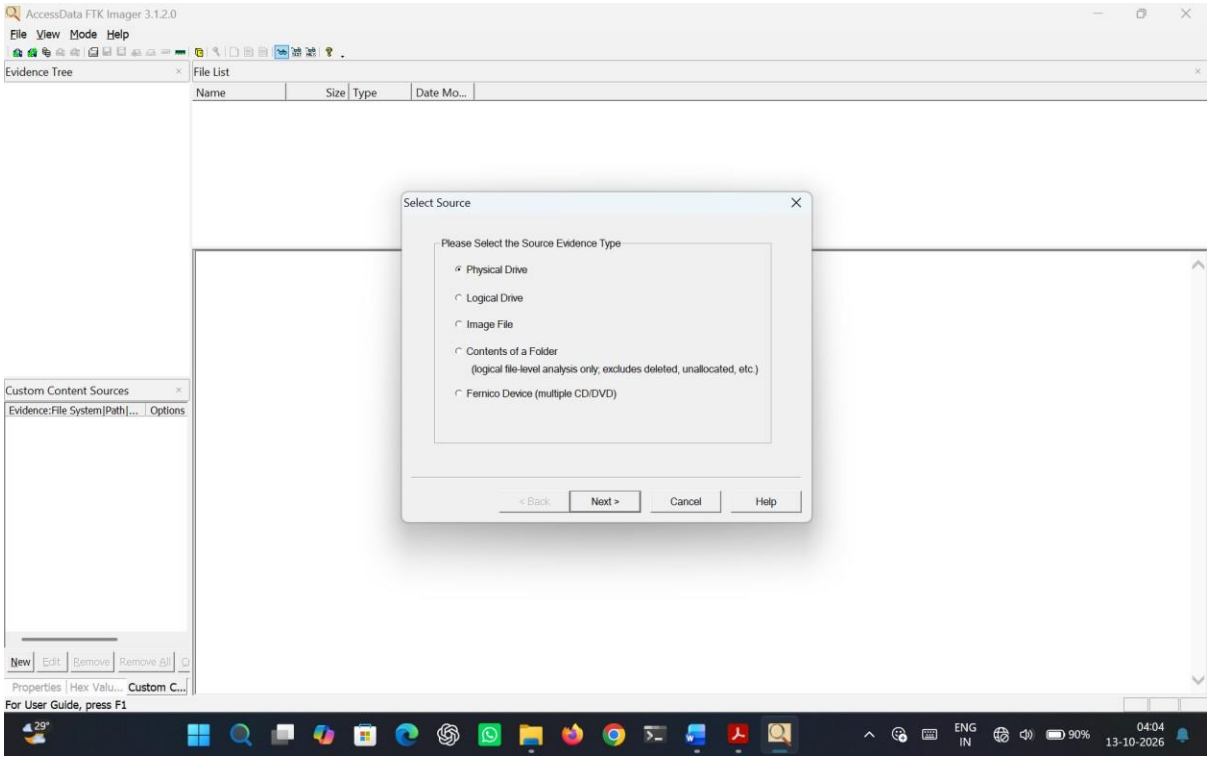


Figure 5.1 Select Source

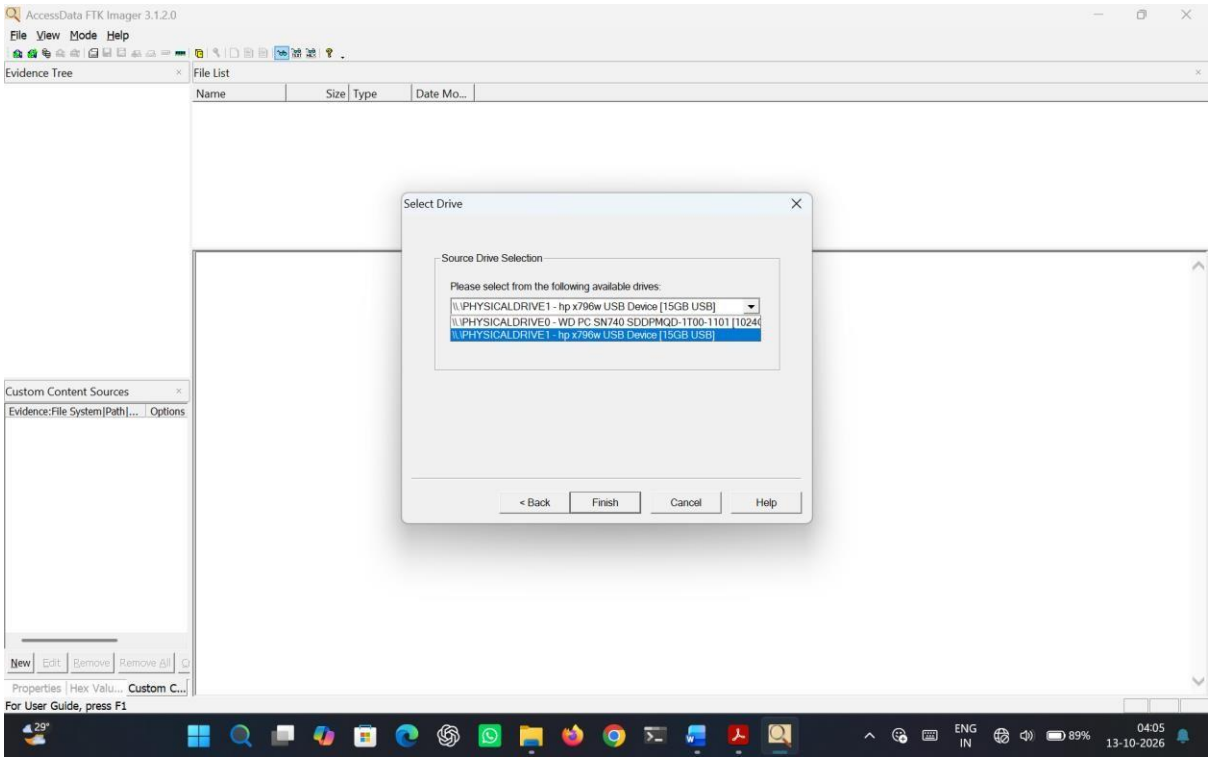


Figure 5.2 Select Drive

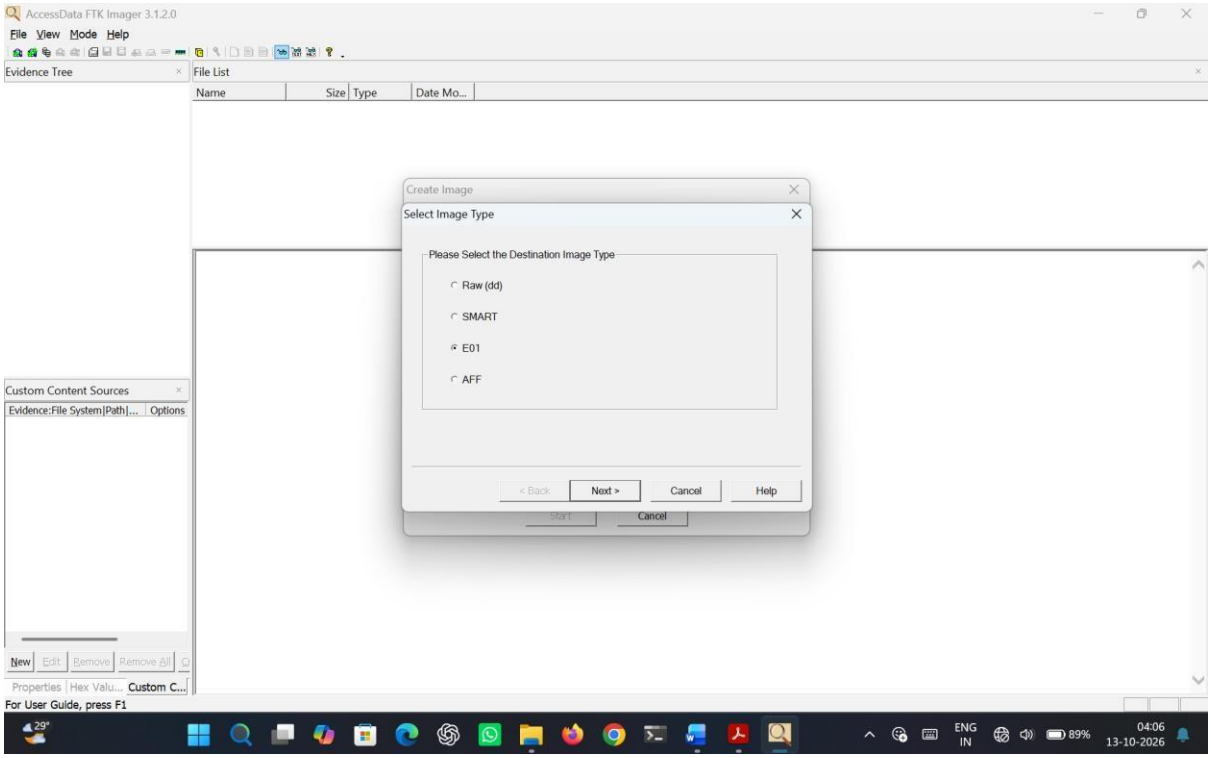


Figure 5.3 Select Image type

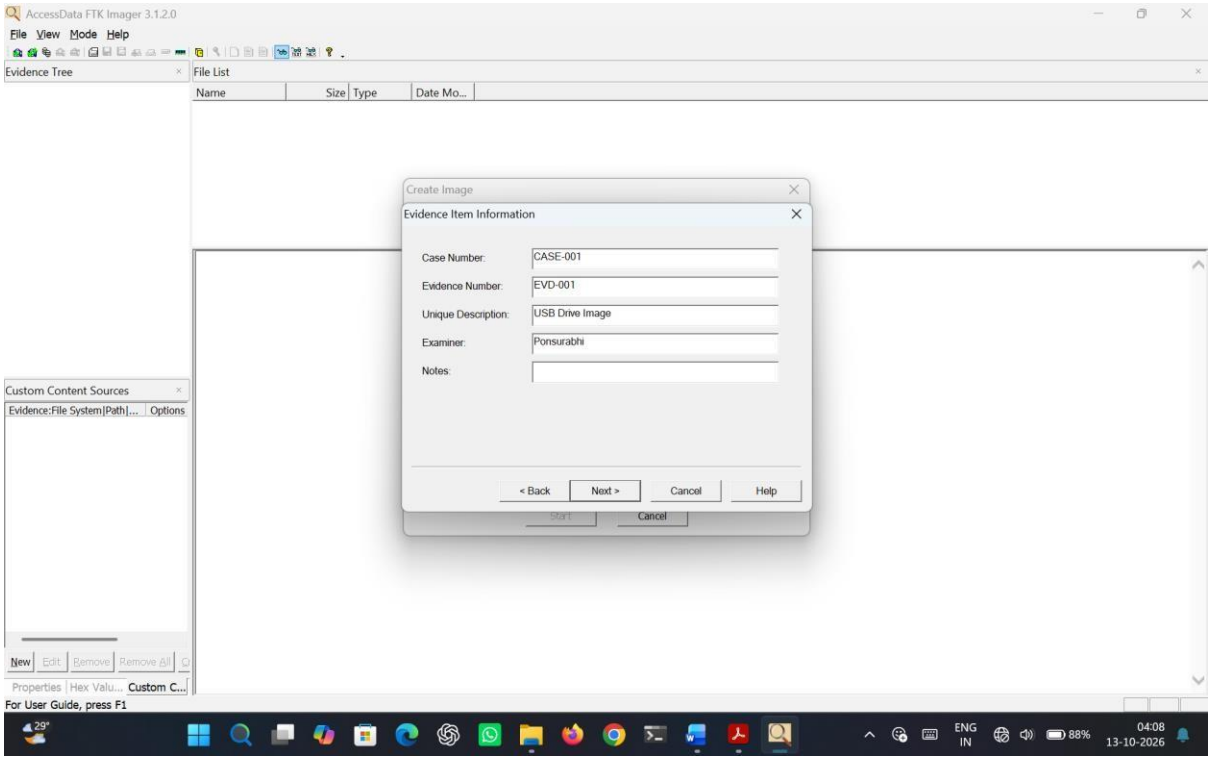


Figure 5.4 Fill evidence item information

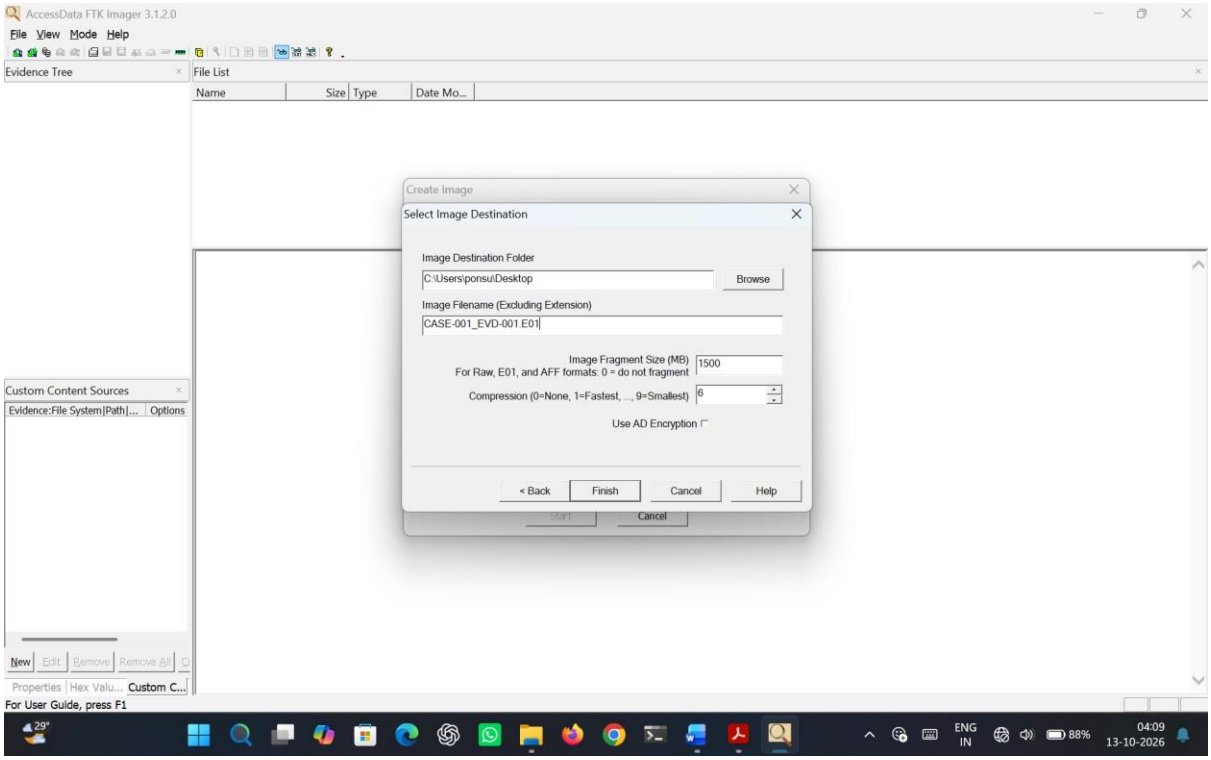


Figure 5.5 Select Image destination

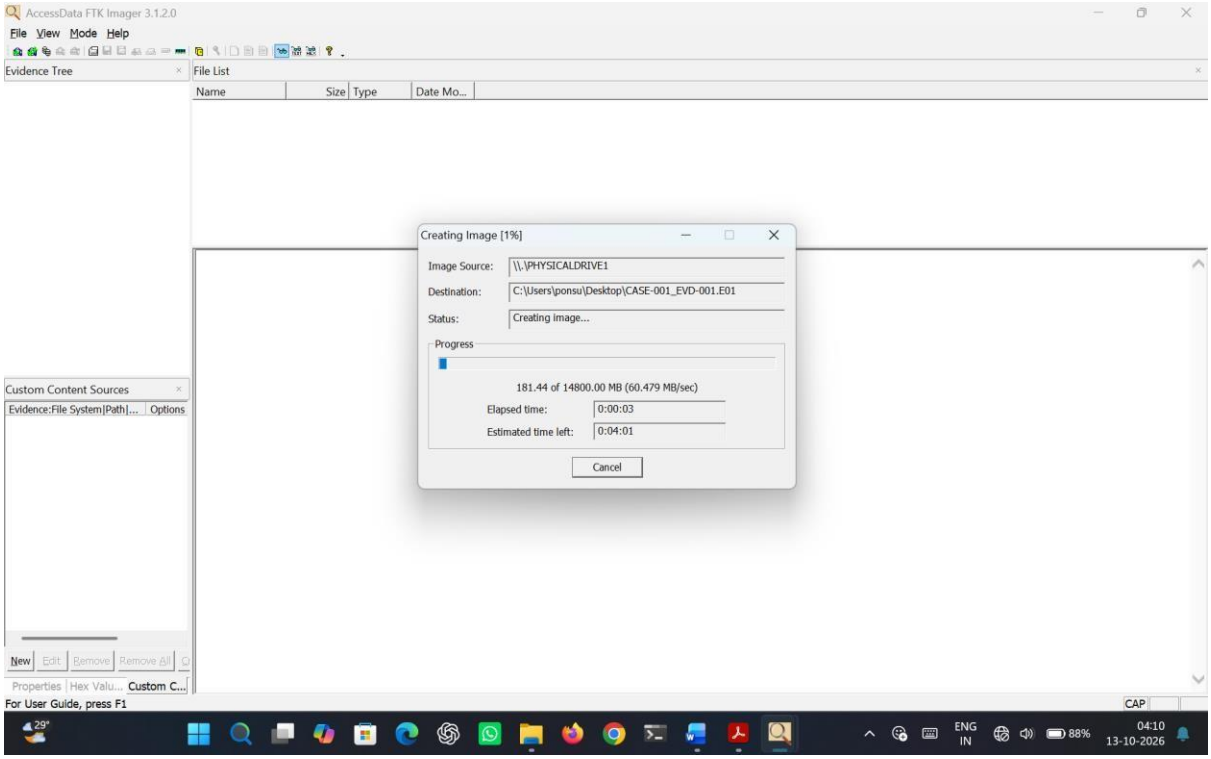


Figure 5.6 Creating Image

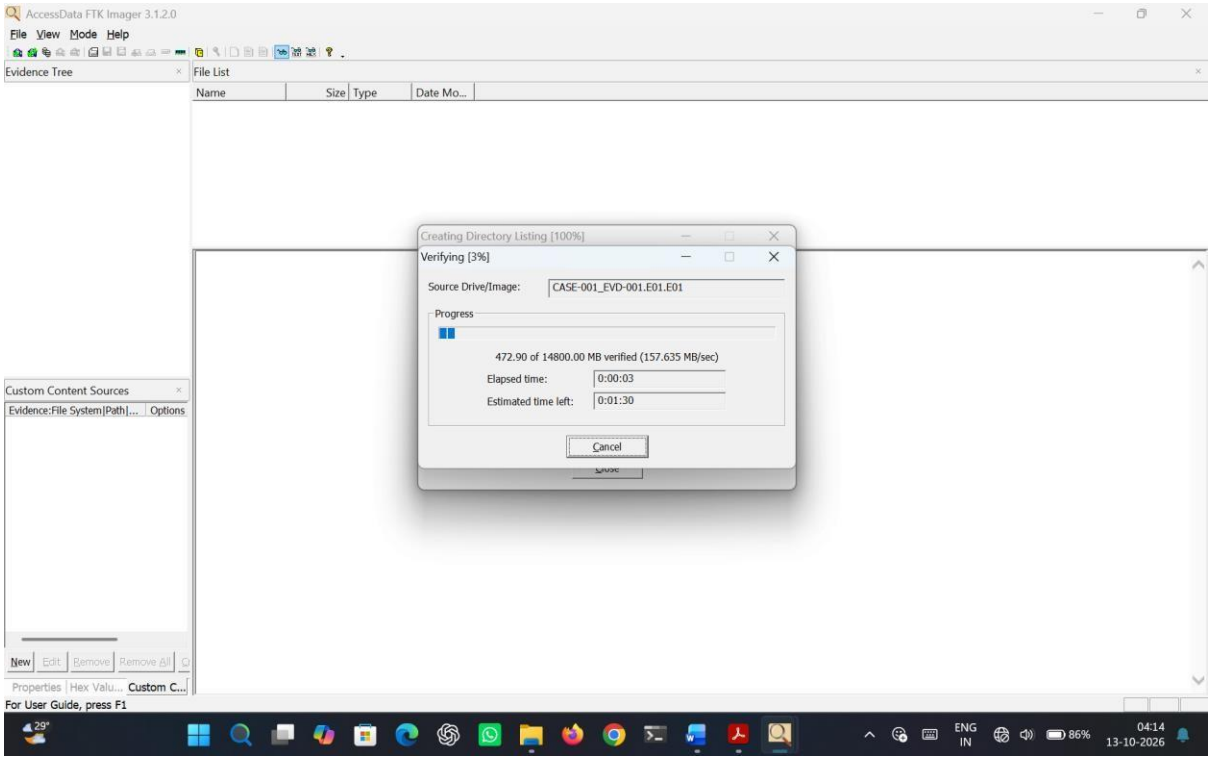


Figure 5.7 Verifying Image

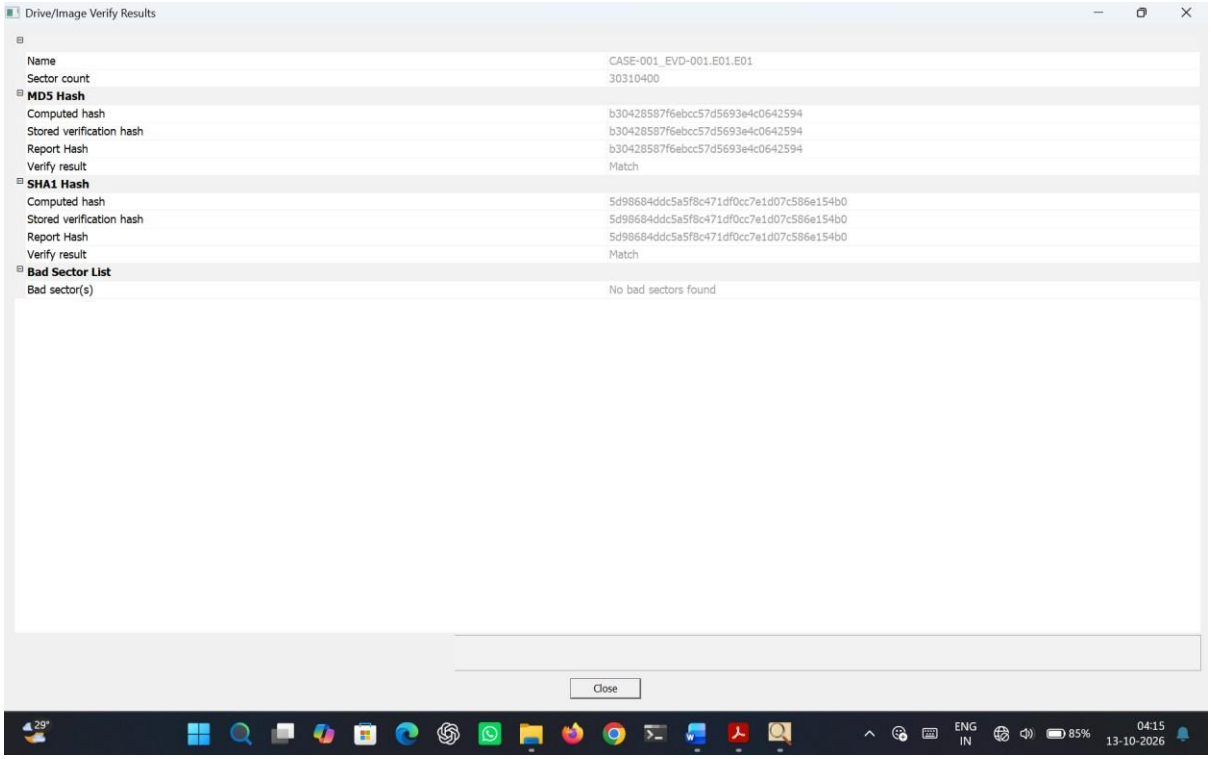


Figure 5.8 Hash generated

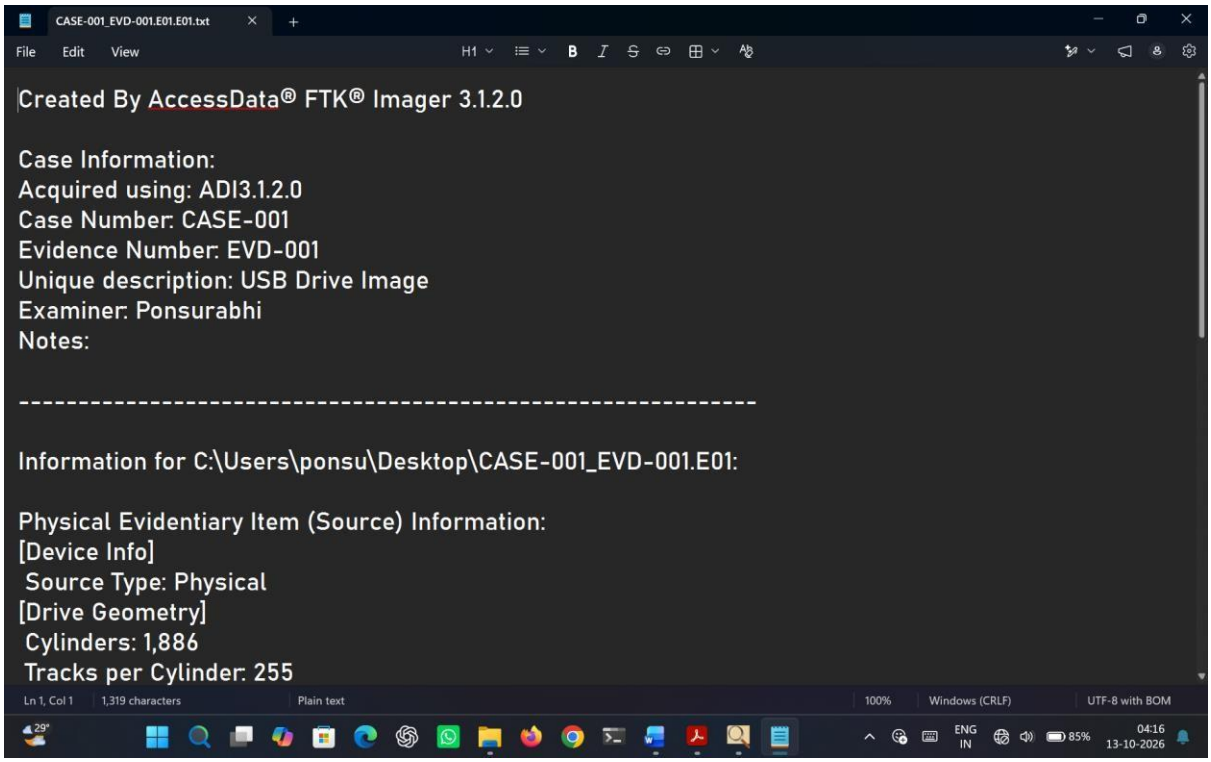


Figure 5.9 Text Report

RESULT

A forensic image of the storage device was successfully acquired and verified using FTK Imager. The generated hash values confirmed that the acquired image maintained data integrity.

EXPERIMENT NO: 6 **DATE: 13/08/2026**

FIND LAST CONNECTED USB ON YOUR SYSTEM

(USB FORENSICS)

AIM

To identify the last connected USB device on your system using USBDeview.

PROCEDURE

1. Download USBDeview

- Open your browser and go to the NirSoft website.
- Download USBDeview (ZIP file).

2. Extract the Downloaded File

- Extract the downloaded ZIP file to any folder.
- Run USBDeview.exe.

3. Run USBDeview as Administrator

- Right-click on USBDeview.exe.
- Select Run as administrator.

4. USBDeview Main Window

- USBDeview lists all USB devices currently connected and previously used.
- Each entry contains detailed information about the USB device.
- The Last Plug/Unplug Date column shows when the USB device was last connected or disconnected.

5. Identify Last Connected USB

- Click on the Last Plug/Unplug Date column header to sort the entries.
- The top entry will show the most recently used USB device.
- Note the following details:
    1. Device Name
    2. Description

3. Serial Number
4. Last Plug/Unplug Date

6. View Detailed Information

- Double-click on any USB device to view detailed information.
- The information includes:
  1. Property
  2. Device Name
  3. Device Type
  4. Serial Number
  5. Last Plug/Unplug Date
  6. Vendor ID
  7. Product ID
  8. USB Class
  9. USB SubClass
  10. USB Protocol

7. Export the Report (Optional)

- Go to the File menu.
- Select the required export option.
- Reports can be saved in formats such as:
  1. HTML
  2. CSV
  3. TXT
- Save the report for documentation.

8. Optional – Uninstall / Disable USB Device

- Right-click on any previously used USB device.
- You can:
  1. Uninstall the selected device.
  2. Disable the selected device.
  3. Enable the selected device.
- Caution: Do not uninstall or disable devices during forensic examination unless specifically required.

9. Optional – Remote Computer

- USBDeview can also be used on a remote computer.
- Log in using an administrator account and run the tool.

10. Record Observations

- Record the last connected USB device details in the lab manual:

OBSERVATION:

1. Device Name: 0000.0014.0000.0003.000.000.000.000.000
2. Description: USB Audio Device
3. Serial Number: Not available
4. Last Plug/Unplug Date: 11-10-2026 05:53:48 AM
5. Vendor ID: 001F
6. Product ID: 0B21

SCREENSHOTS:

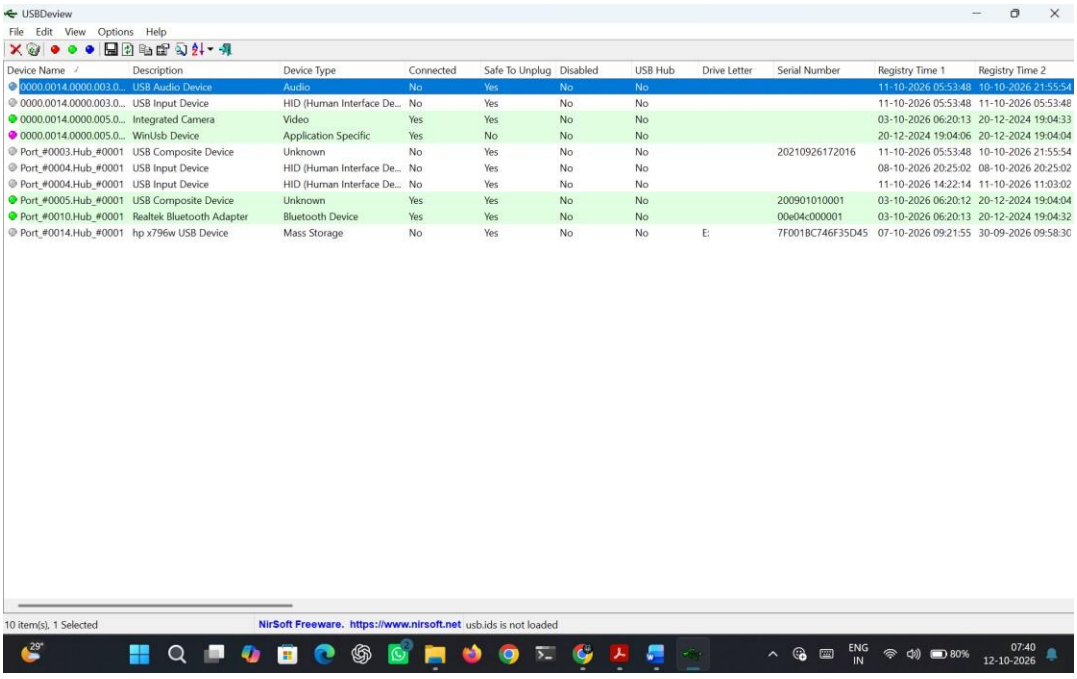


Figure 6.1 Results of USBDeview

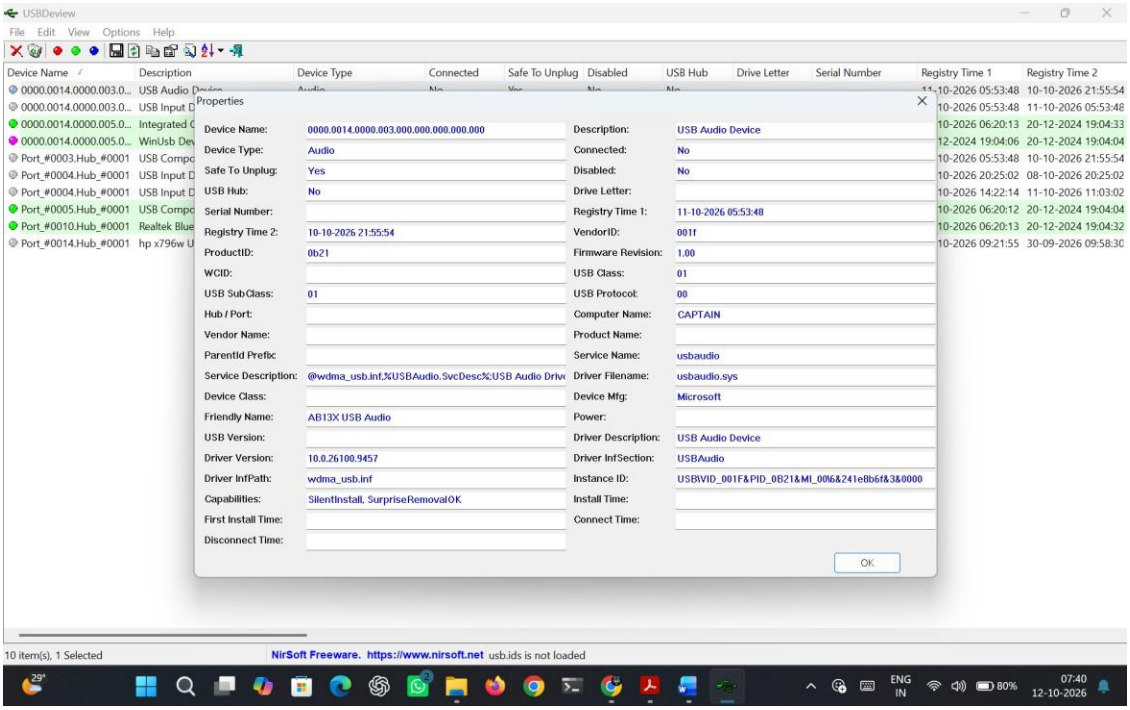
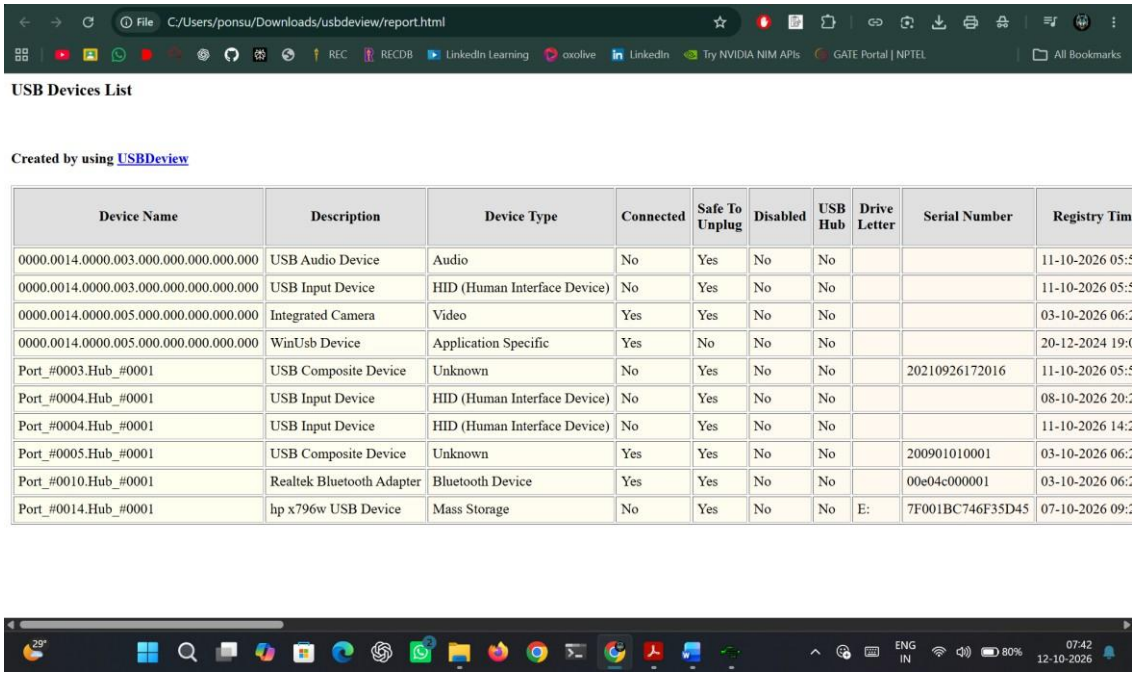


Figure 6.2 View Detailed Information



Device Name	Description	Device Type	Connected	Safe To Unplug	Disabled	USB Hub	Drive Letter	Serial Number	Registry Tim
0000.0014.0000.003.000.000.000.000.000	USB Audio Device	Audio	No	Yes	No	No			11-10-2026 05:5
0000.0014.0000.003.000.000.000.000.000	USB Input Device	HID (Human Interface Device)	No	Yes	No	No			11-10-2026 05:5
0000.0014.0000.005.000.000.000.000.000	Integrated Camera	Video	Yes	Yes	No	No			03-10-2026 06:2
0000.0014.0000.005.000.000.000.000.000	WinUsb Device	Application Specific	Yes	No	No	No			20-12-2024 19:(
Port_#0003.Hub_#0001	USB Composite Device	Unknown	No	Yes	No	No		20210926172016	11-10-2026 05:5
Port_#0004.Hub_#0001	USB Input Device	HID (Human Interface Device)	No	Yes	No	No			08-10-2026 20:2
Port_#0004.Hub_#0001	USB Input Device	HID (Human Interface Device)	No	Yes	No	No			11-10-2026 14:2
Port_#0005.Hub_#0001	USB Composite Device	Unknown	Yes	Yes	No	No		200901010001	03-10-2026 06:2
Port_#0010.Hub_#0001	Realtek Bluetooth Adapter	Bluetooth Device	Yes	Yes	No	No		00e04c000001	03-10-2026 06:2
Port_#0014.Hub_#0001	hp x796w USB Device	Mass Storage	No	Yes	No	No	E:	7F001BC746F35D45	07-10-2026 09:2

Figure 6.3 HTML Report

RESULT:

Thus USBDeview was used to identify all USB devices connected to the system, along with the exact date and time of their last connection.

EXPERIMENT NO: 7 **DATE: 20/08/2026**

HOW TO VIEW LAST ACTIVITY OF YOUR PC

AIM

To view the last activity of your PC.

TOOL

LastActivityView (NirSoft)

INTRODUCTION

LastActivityView is a tool for Windows operating systems that collects information from various sources on a running system and displays a log of actions made by the user and events that occurred on the computer.

PROCEDURE

1. Download LastActivityView

- Open your browser and go to the NirSoft website.
- Search for LastActivityView.
- Download LastActivityView.

2. Extract the Downloaded File

- Download the ZIP file.
- Extract the ZIP file using WinRAR/7-Zip to any folder.
- Locate LastActivityView.exe.

3. Run LastActivityView

- Double-click on LastActivityView.exe.
- No installation is required.

4. Explore the Main Window

- LastActivityView displays all recent activities in chronological order.
- You can view:
    1. Action Time

2. Description
3. Filename / Full Path
4. Other activity information

5. Filter Activities (Optional)

- Use the Filter option to view specific types of activities.
- Examples:
  1. Run EXE file
  2. Open file
  3. Delete file
  4. System events

6. View Detailed Information

- Double-click on any activity entry.
- A detailed information window will open with additional information such as:
  1. Action Time
  2. Description
  3. Filename / Full Path
  4. More Information
  5. File Extension
  6. File Type
  7. Product Name
  8. Product Version
  9. Company Name

7. Save Report

- Go to the File menu.
- Save the selected items or all items.
- Available formats include:
  1. HTML
  2. CSV
  3. Text
  4. XML

8. Record Observations

9. Analyze Findings

- Analyze the collected information.
- Correlate the information to understand the last activities performed by the user.

10. Close the Tool

- After analysis, close LastActivityView.
- The tool does not make any changes to the system.

OBSERVATION:

1. Last Executed Program: CONSENT.EXE

Path: C:\WINDOWS\SYSTEM32\CONSENT.EXE

2. Last Opened File: Acrobat.exe

Path: C:\PROGRAM FILES\ADOBE\ACROBAT DC\ACROBAT\ACROBAT.EXE

3. System Start Time: 12-10-2026, 07:26:13

4. Installed Software: Google Chrome, Adobe Acrobat DC, Microsoft Edge WebView2, Lenovo Modern IM Controller, and Windows Phone Link.

5. Any Other Activity: Execution of system utilities such as SEARCHFILTERHOST.EXE, SEARCHPROTOCOLHOST.EXE, and CONHOST.EXE was recorded. Multiple executions of Chrome.exe and MSEDGEWEBVIEW2.EXE were also observed.

SCREENSHOTS:

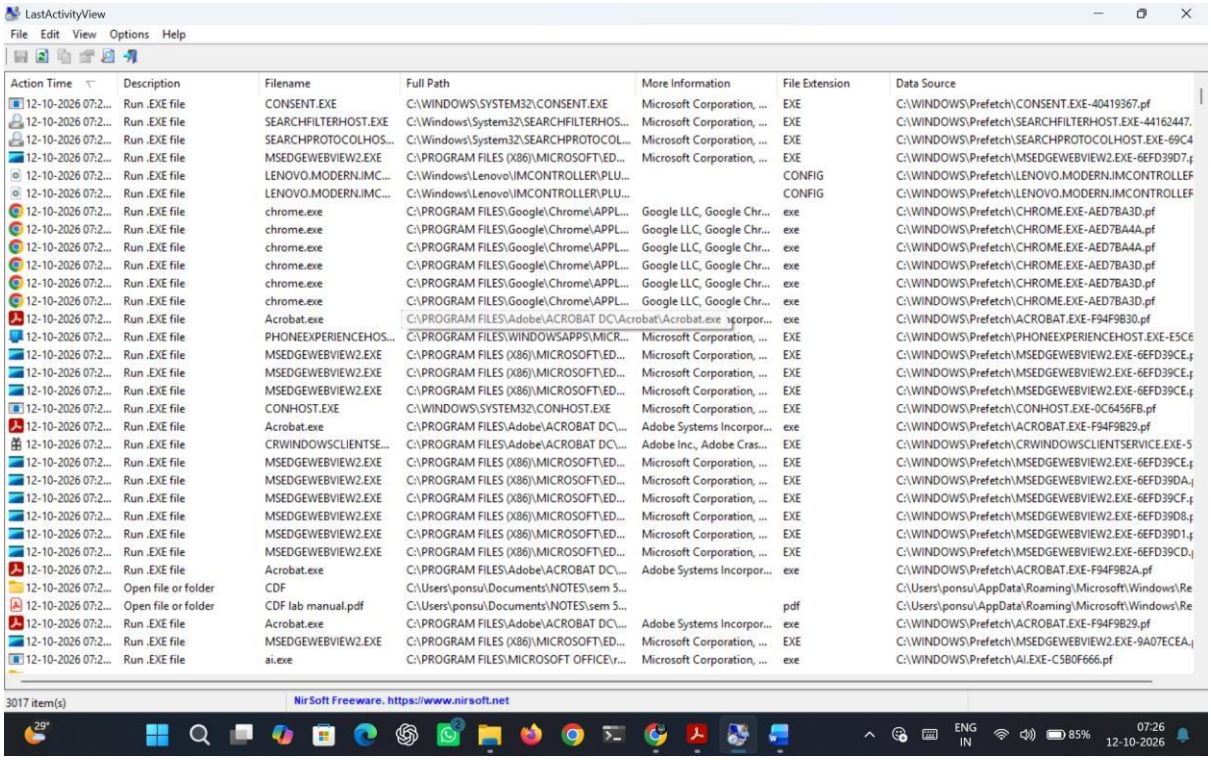


Figure 7.1 LastActivityView Results

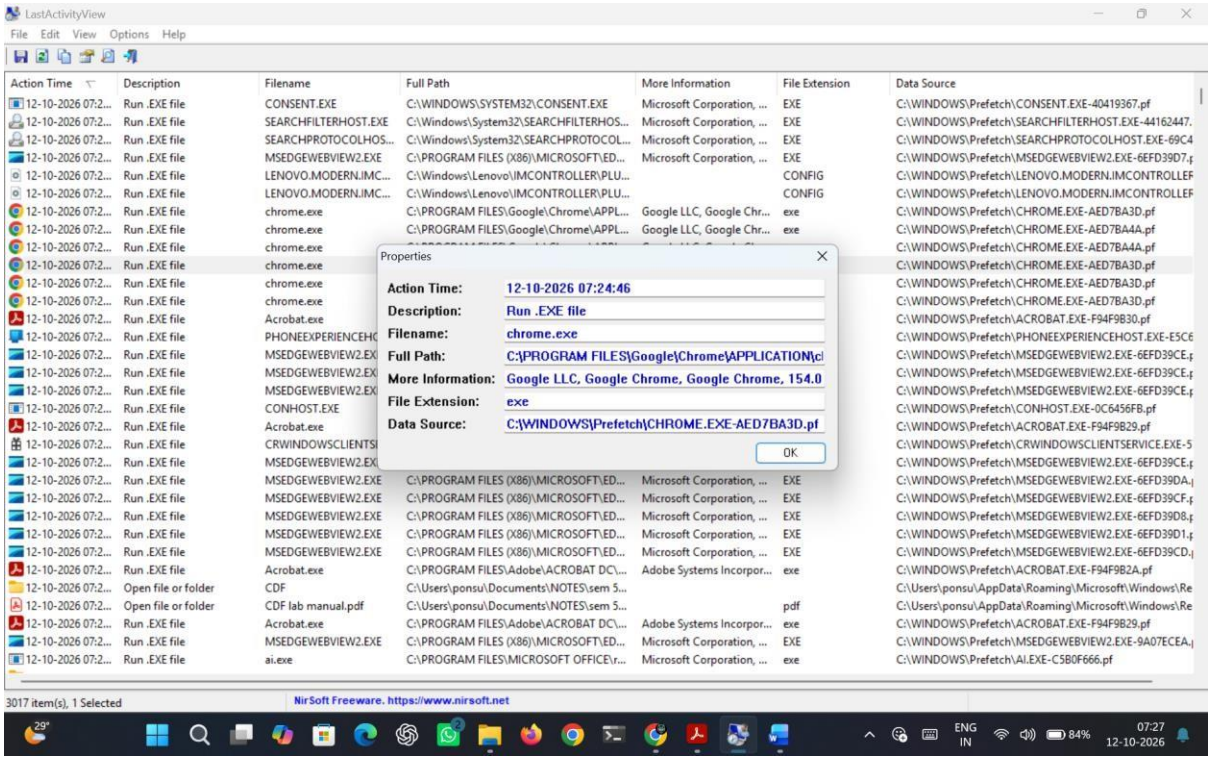


Figure 7.2 Properties of one entry

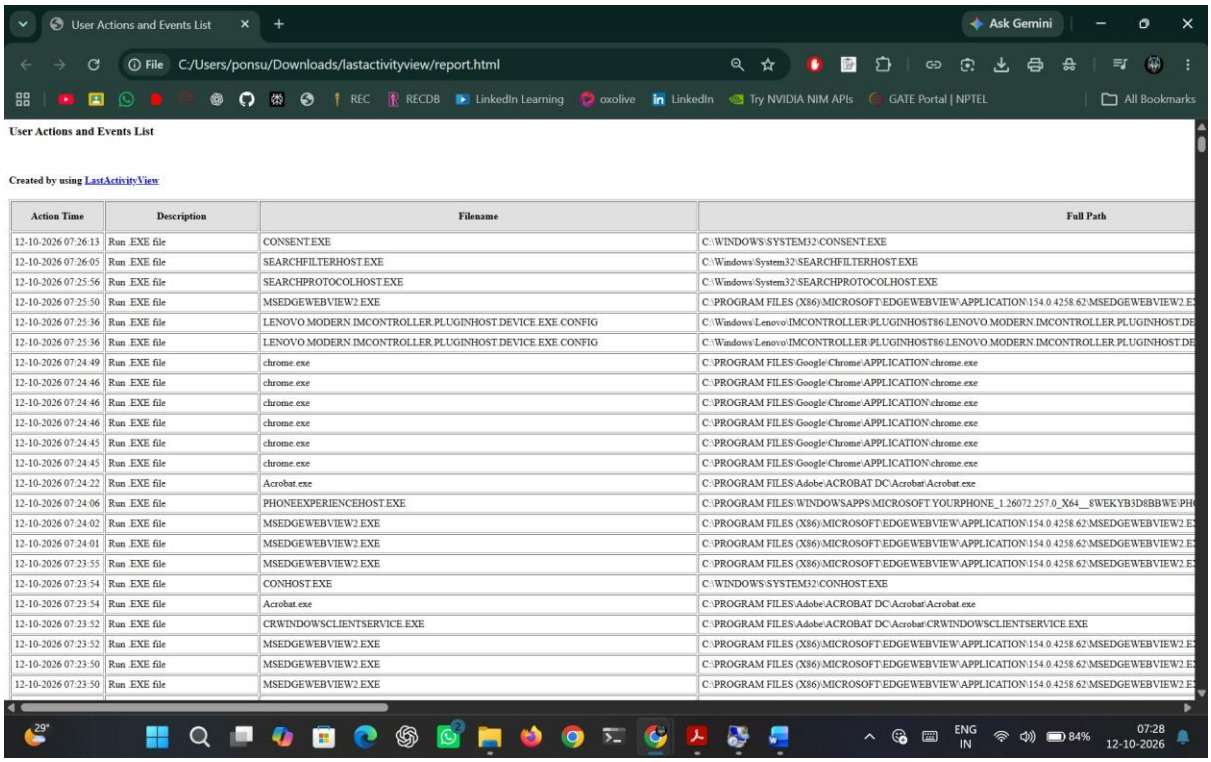


Figure 7.3 HTML Report

RESULT

LastActivityView was used to collect and view detailed logs of user activities and system events that occurred on a Windows PC.

EXPERIMENT NO: 8 **DATE:27/8/2026**

HOW TO EXTRACT BROWSER ARTIFACTS

AIM

To extract browser artifacts using ChromeHistoryView.

TOOL

ChromeHistoryView (NirSoft)

ChromeHistoryView is a small utility that reads the history data file of the Google Chrome web browser and displays the list of all visited web pages in the last days.

Information displayed:

- URL
- Title
- Visit Date/Time
- Number of Visits
- Number of times that the user typed this address (Typed Count)
- Referrer
- Visit ID

PROCEDURE

Step 1: Download ChromeHistoryView

1. Open your browser and go to the NirSoft website.
2. Search for ChromeHistoryView.
3. Download the tool from the official website.

Step 2: Extract the Downloaded File

1. Download the ZIP file.
2. Extract it using WinRAR or 7-Zip to any folder.
3. Locate the extracted files, including ChromeHistoryView.exe.

Step 3: Run ChromeHistoryView

1. Double-click on ChromeHistoryView.exe.
2. No installation is required.
3. The tool will automatically load your Chrome history.

Step 4: Explore the Main Window

1. The main window lists all visited web pages.
2. Each entry shows detailed information about the visit.

Step 5: View Additional Columns

1. Click the View menu.
2. Select the columns you want to display.

Step 6: Understand the Columns

URL: Web address visited.

Title: Title of the webpage.

Visit Time: Date and time of the visit.

Visit Count: Number of times the webpage was visited.

Typed Count: Number of times the user typed the URL.

Referrer: Website from which the link was opened.

Visit ID: Unique ID for the visit.

Step 7: Export the Data (Optional)

1. Go to the File menu and select the export option.
2. Save the data in CSV, HTML, or text format for further analysis.

Step 8: Analyze the Artifacts

1. Review the list of visited websites.
2. Identify interesting web activities, like frequently visited sites, search engines, social media, emails, etc.

Step 9: Record Observations

1. Note important details in your lab manual.
2. Capture findings such as the most visited sites, suspicious URLs, timestamps, and referrers.

Step 10: Close the Tool

1. After analysis, close ChromeHistoryView.
2. The tool does not make any changes to the system or browser data.

OBSERVATION:

1. **Most Visited Site:** https://dev.azure.com/ponsurabhi365/BookSphere
2. **Total Number of Visits:** 5,242 items.
3. **Suspicious URL (if any):** https://www.google.com/search?q=shaft+collar
4. **Date and Time:** 02-10-2026, 04:33 AM to 09:26 AM.
5. **Referrer Website:** https://www.google.com/ and https://dev.azure.com/ponsurabhi365/BookSphere
6. **Remarks:** Browser history was successfully extracted and analyzed using ChromeHistoryView. The visited URLs, webpage titles, visit dates and times, visit counts, referrers, and visit durations were displayed for digital forensic analysis.

SCREENSHOTS:

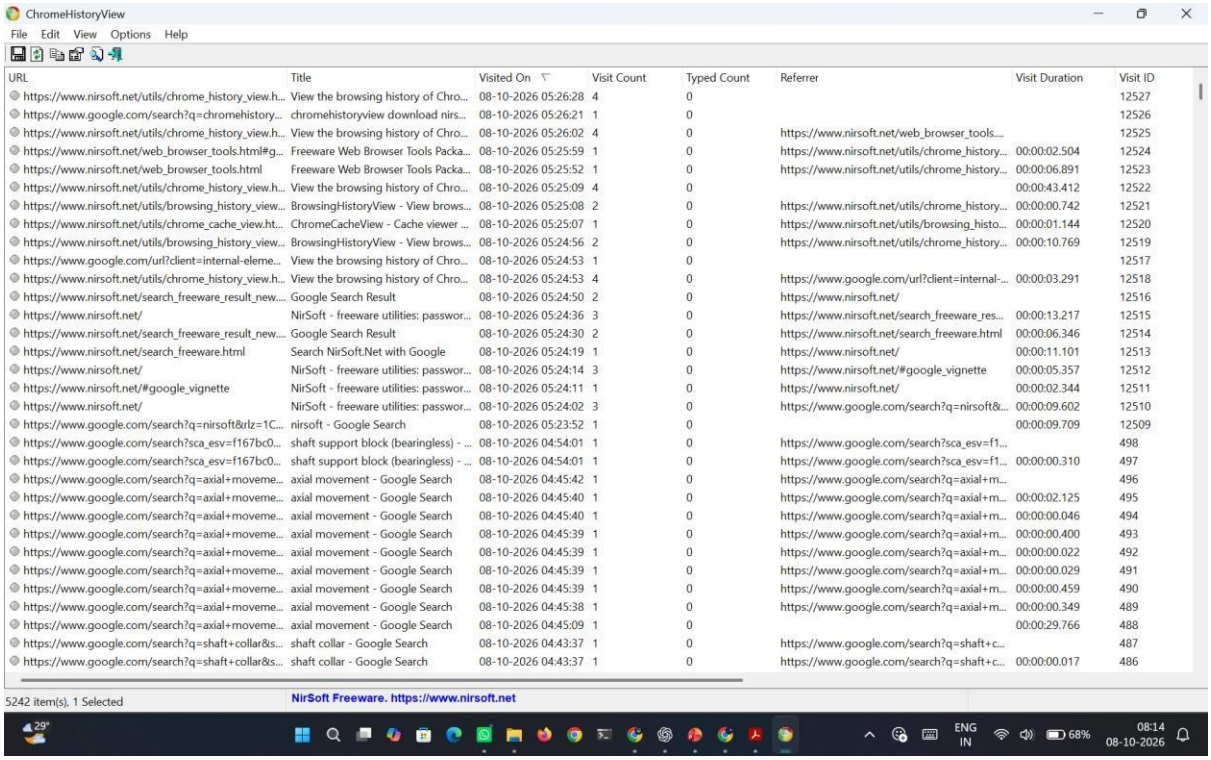


Figure 8.1 Chrome History View Results

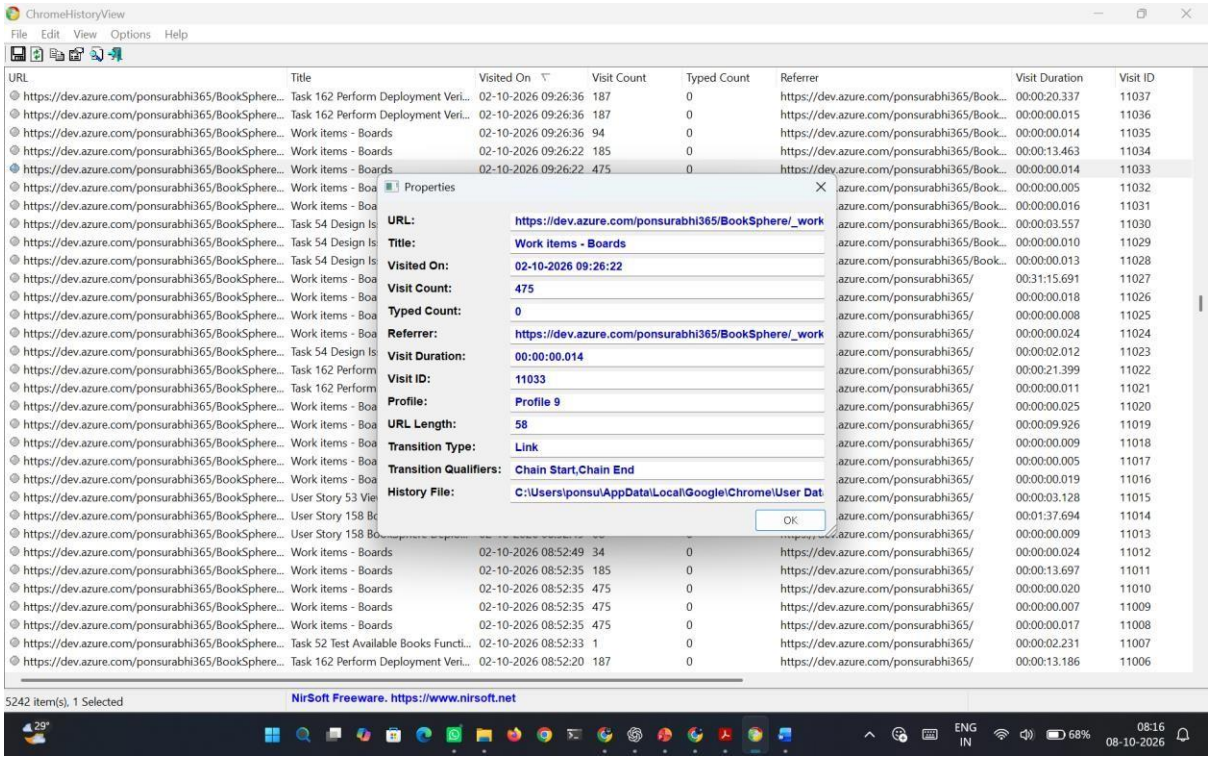


Figure 8.2 Detailed Information of a single entry

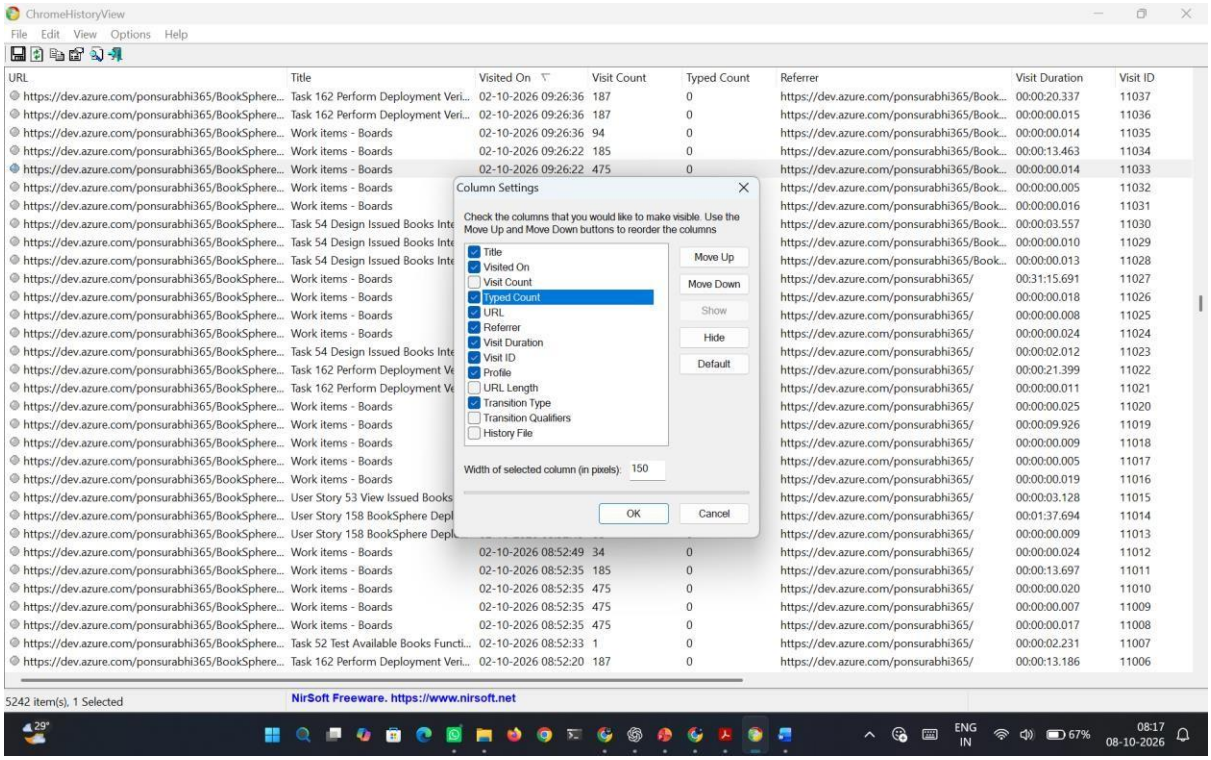


Figure 8.3 Adjust Column Settings

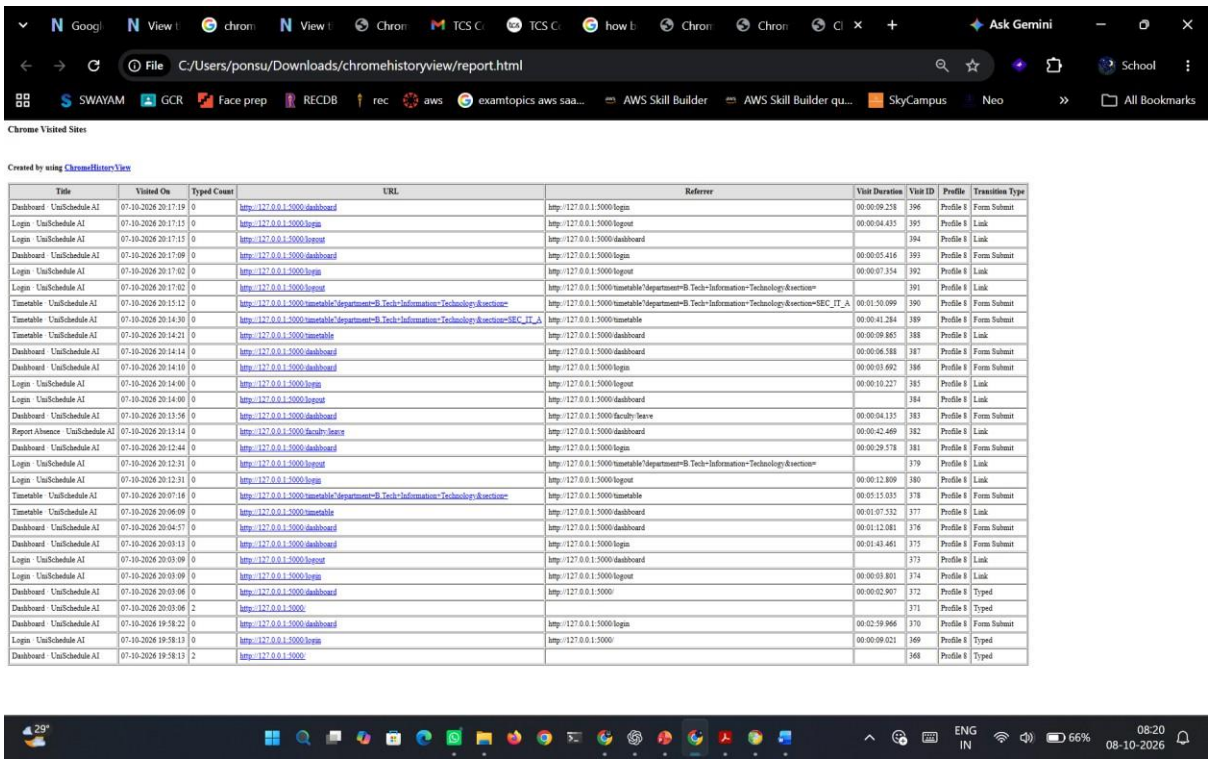


Figure 8.4 HTML Report

RESULT:

Thus ChromeHistoryView was used to extract and analyze browser history.

EXPERIMENT NO: 9 **DATE: 10/09/2026**

COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE

AIM

To compare two files for forensic investigation using Compare It! software.

ABOUT COMPARE IT

Compare It is a software tool that displays two files side-by-side, with colored differences to simplify analysis. It can compare changes between files with a single mouse click or keystroke. It also allows users to edit files directly in the comparison window and provides a colored printout of the differences report.

PROCEDURE

1. Install Compare It Software

- Download Compare It! from the official source.
- Install Compare It! on the system.

2. Launch Compare It

- Click on the Compare It! icon on the desktop.

3. Select First File

- In the Compare window, click on the left-side Folder icon.
- Select the first file.
- Click Open.

4. Select Second File

- Click on the right-side Folder icon.
- Select the second file.
- Click Open.

5. Files Loaded for Comparison

- Both files will be loaded into the Compare window.
- Click Compare to start the comparison.

6. View Comparison Result

- The two files will be displayed side-by-side with colored differences.
- The differences are highlighted using different colors.

7. Analyze Differences

- Review the highlighted differences carefully.
- Color indicators:

1. Green – Added content
2. Red – Deleted content
3. Yellow – Modified content

- The toolbar or keyboard shortcuts can be used to navigate through the differences.

8. Edit (Optional)

- The files can be edited directly in the comparison window.
- Save the changes if required.

9. Generate Report (Optional)

- Click the Print/Report option to generate a colored differences report.
- Print or save the report for documentation.

10. Close the Tool

- After completing the analysis, close Compare It!

OBSERVATION:

Statistics:

Total changed lines: 1 (<1%) Same: 466

Target Only: 0

Source Only: 0

Changed: 1

SCREENSHOTS:

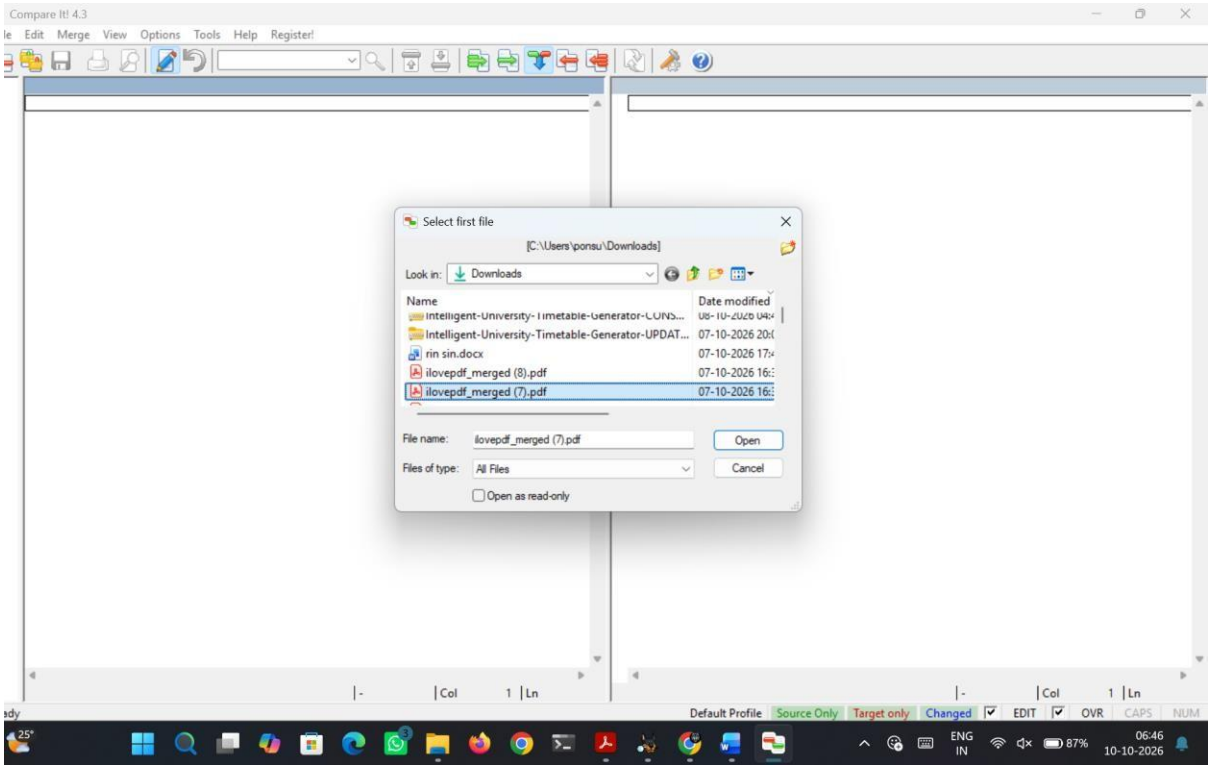


Figure 9.1 Load first file

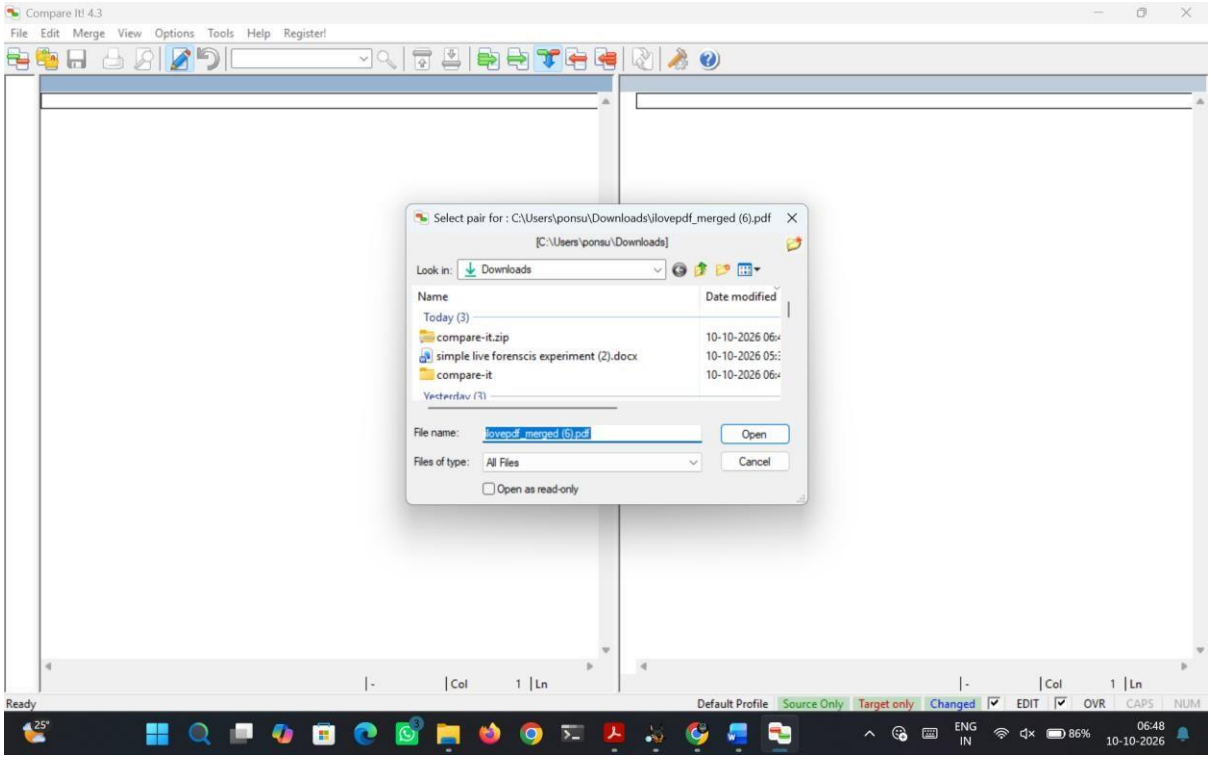


Figure 9.2 Load Second File

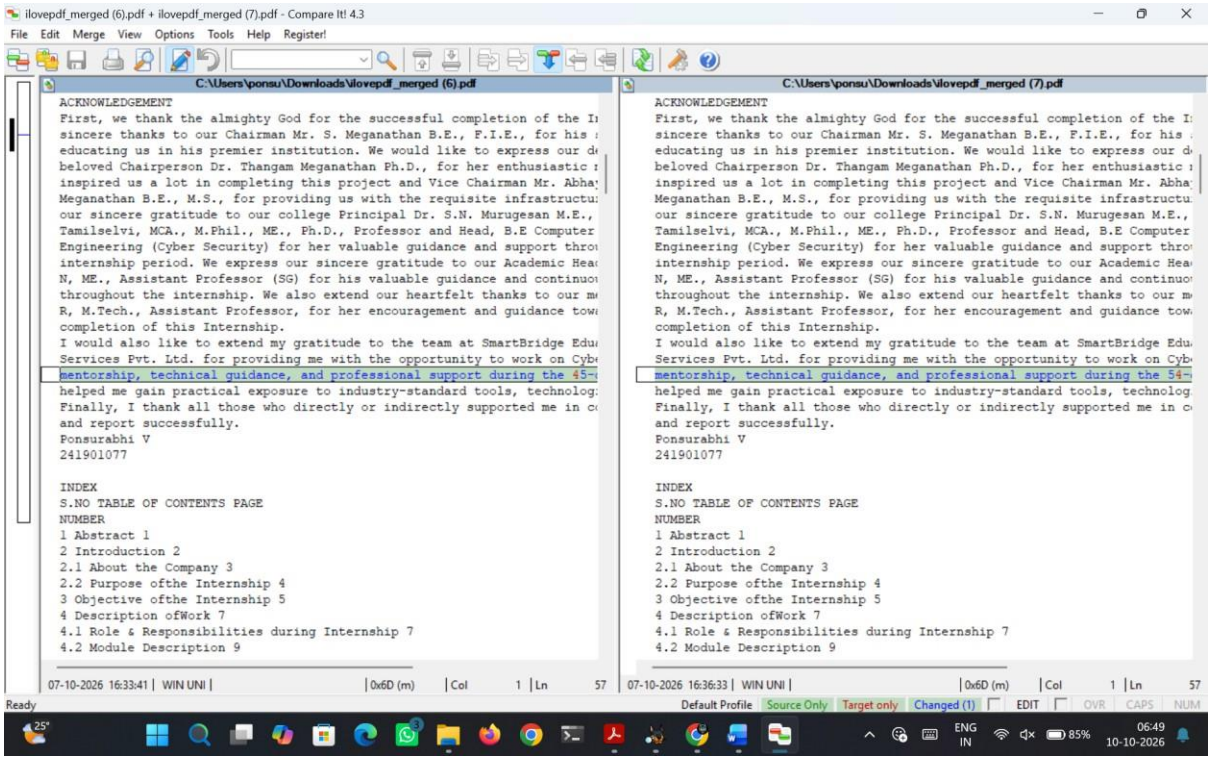


Figure 9.3 Comparison of two files

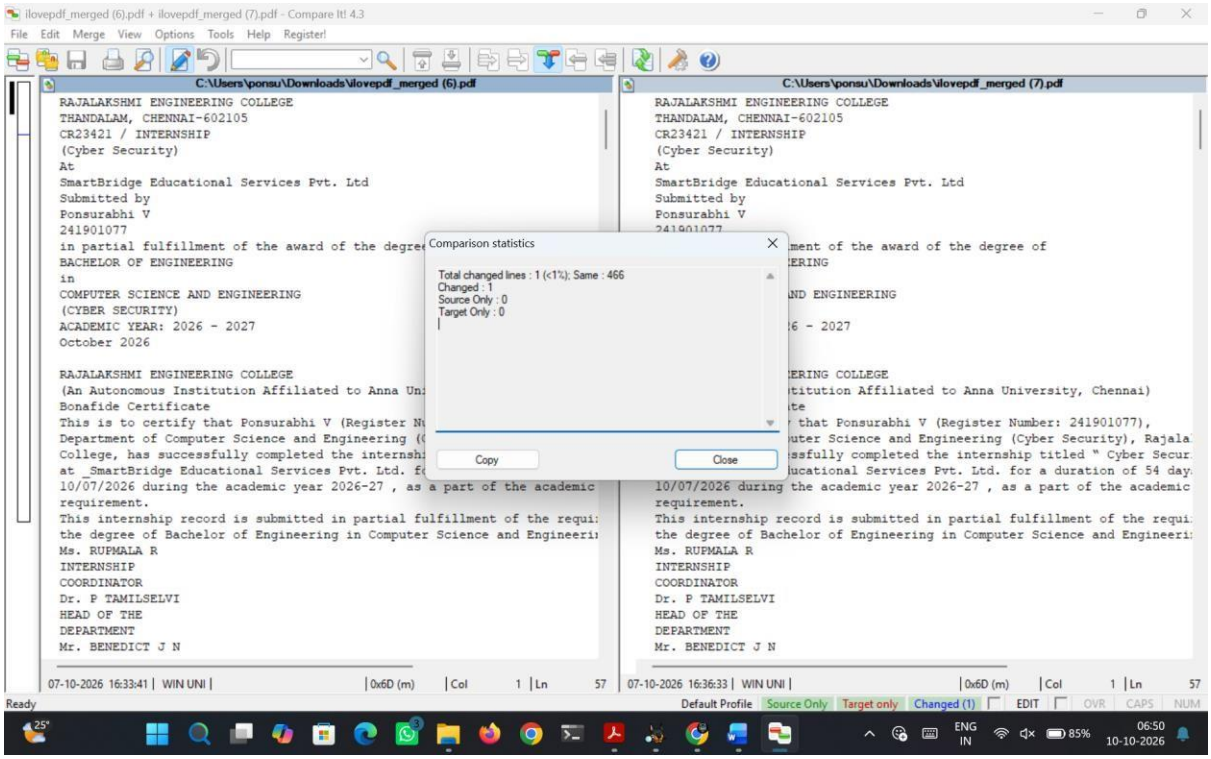


Figure 9.4 Comparison Statistics

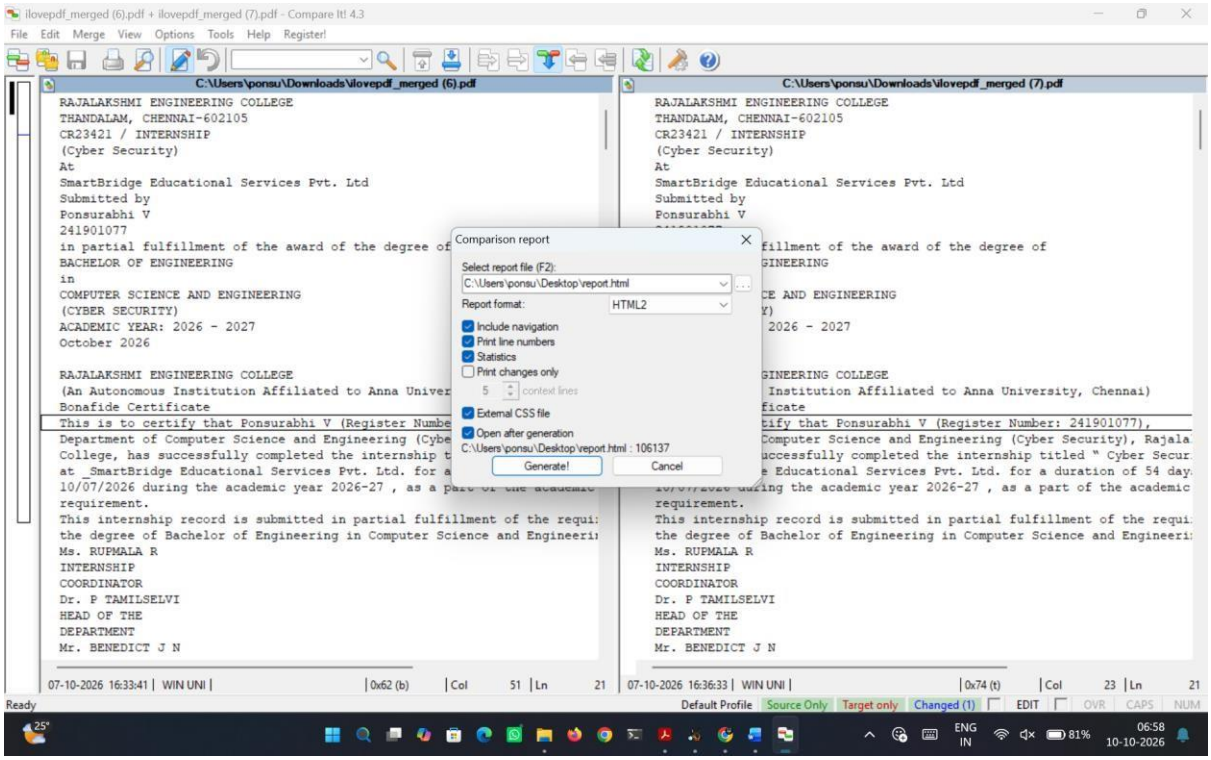


Figure 9.5 Generate Report

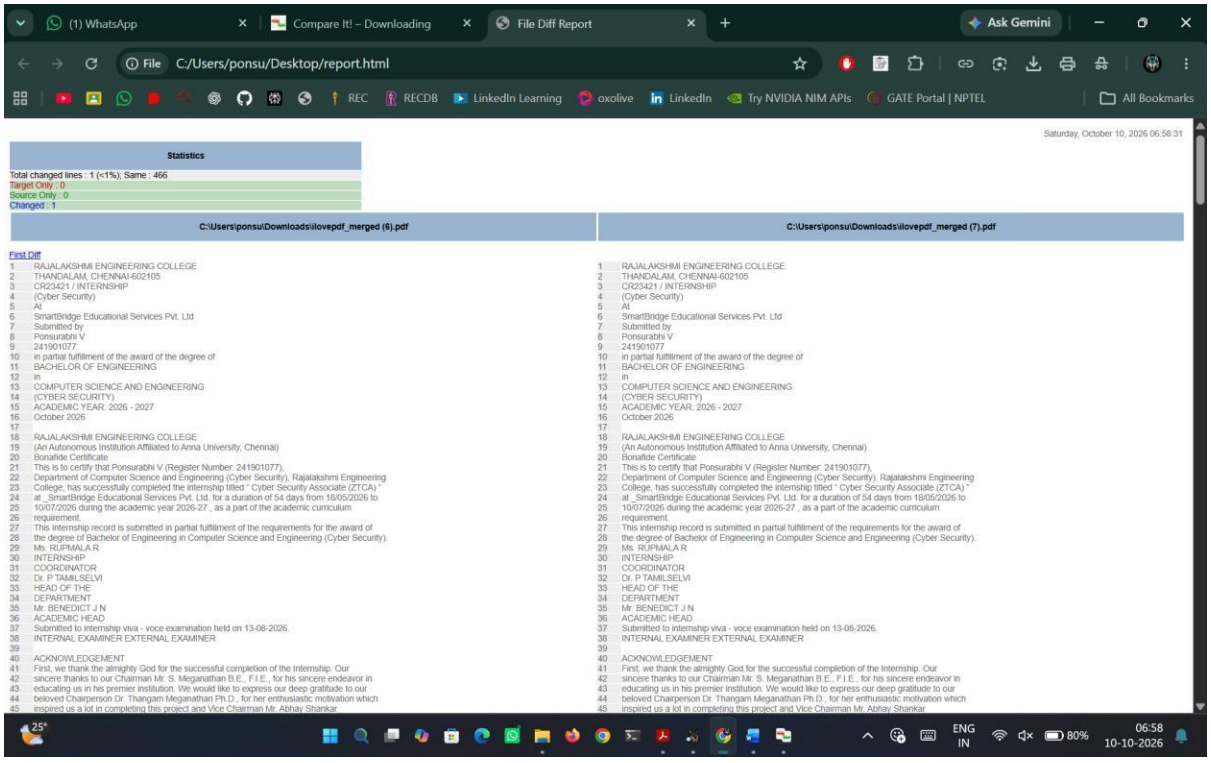


Figure 9.6 HTML Report

RESULT

Thus Compare It was used to identify differences between two files quickly and accurately.

EXPERIMENT NO: 10 **DATE: 17/09/2026**

HOW TO COLLECT EMAIL EVIDENCE IN VICTIM PC

AIM

To collect email evidence from a victim PC.

TOOL

DumpIt (Forensic Memory Dumper)

ABOUT DUMPIT

DumpIt is a forensic memory acquisition tool used to dump the physical memory (RAM) of a Windows machine. It works on x86 (32-bit) and x64 (64-bit) Windows systems and is useful for forensic analysis and email artifact extraction.

PROCEDURE

1. Download DumpIt

- Open your browser and go to the MoonSols DumpIt website.
- Download the latest version of DumpIt.

2. Extract the Downloaded File

- Extract the ZIP file to any folder.
- The extracted folder will contain DumpIt.exe.

3. Run DumpIt as Administrator

- Right-click on DumpIt.exe.
- Select Run as administrator.

4. Confirm Memory Dump

- A confirmation message will appear.
- Click Yes to start capturing the system RAM.

5. Memory Dump in Progress

- DumpIt will start capturing the physical memory.

- A progress indicator will show the status of the acquisition.

6. Dump Completed

- Once the dump is complete, a success message will be displayed.
- The memory dump file will be created.

7. Locate the Dump File

- The raw memory dump file memory.raw will be saved in the same directory where DumpIt.exe is located.
- The file size will be approximately equal to the installed RAM size.

8. Preserve the Evidence

- Copy the memory.raw file to a secure external drive.
- Do not modify or rename the evidence file.

9. Analyze the Memory Dump

- Use memory forensic tools such as Volatility or Rekall to analyze the RAM dump.
- Extract relevant email artifacts, including:
    1. Email addresses
    2. Credentials
    3. Email-related data
    4. Communication artifacts
    5. Other relevant information

10. Document the Process

- Record important details such as:
    1. Date & Time of collection
    2. System information
    3. DumpIt version
    4. File name and size
    5. MD5/SHA hash values
    6. Chain of custody

11. Close & Secure

- Close the DumpIt window.
- Ensure the original system remains unchanged.
- Secure the collected memory dump as forensic evidence.

OBSERVATION:

1. Date & Time: 17-09-2026, 14:10:11 AM
2. System Info: Windows Desktop (DESKTOP-6JB69KD)
3. DumpIt Version: v1.3.2.20110401
4. File Name: DESKTOP-6JB69KD-20260917-081011.raw
5. File Size: 35,068 MB (approximately 34.25 GB)
6. Collected By: Ponsurabhi V

SCREENSHOTS:

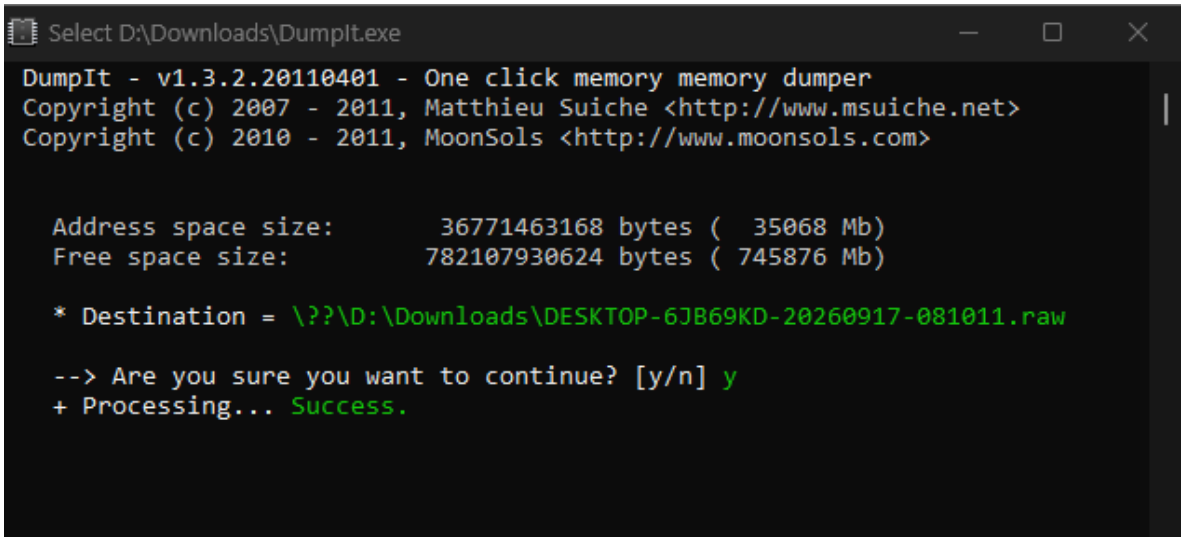


Figure 10.1 Run DumpIt Tool

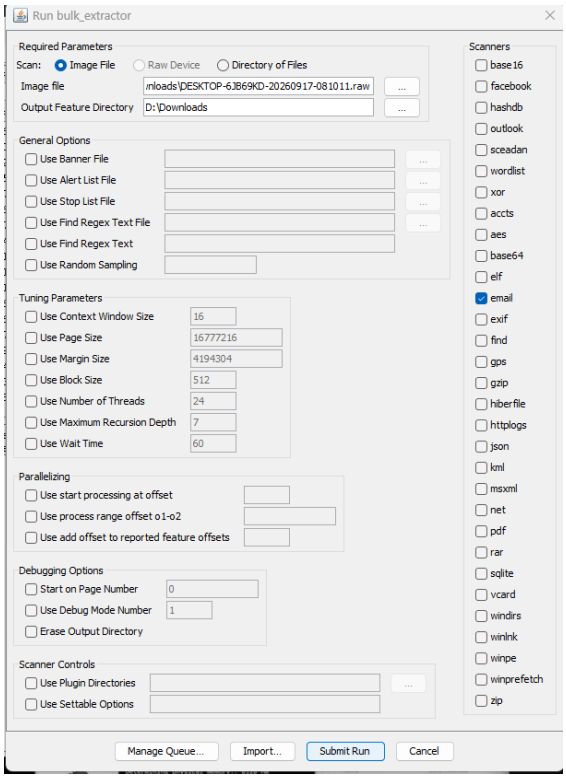


Figure 10.2 Run Bulk Extractor

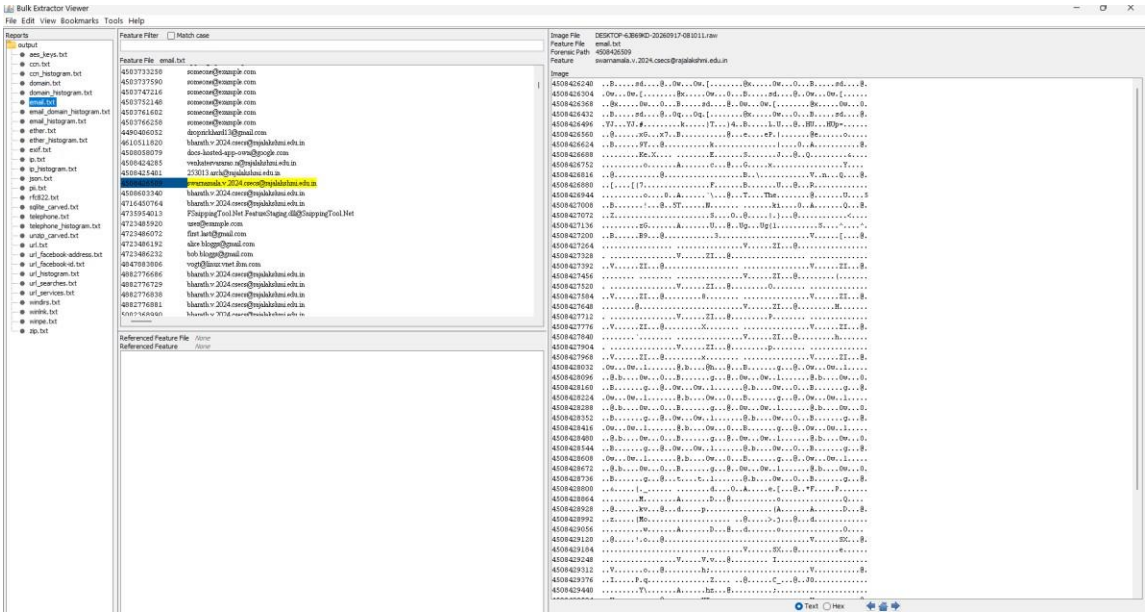


Figure 10.3 View Dumped Memory

RESULT

DumpIt was used capture volatile data (RAM) from a victim machine successfully.

EXPERIMENT NO: 11 **DATE: 24/09/2026**

HIDE AND EXTRACT ANY TEXT FILE BEHIND AN IMAGE FILE USING COMMAND PROMPT

AIM

To study the steps for hiding and extracting any text file behind an image file using Command Prompt.

TOOLS

Windows Command Prompt

CONCEPT

- Any file such as .txt, .rar, .jpg, etc. can be merged inside another file.
- A text file (A.txt) is hidden inside an image file (B.jpg).
- A new file (C.jpg) is created.
- Later, the hidden text file can be extracted from C.jpg.

PROCEDURE

PART 1 – HOW TO HIDE THE FILE

1. Prepare the Files

- Keep the following files in the same folder:
- A.txt → Text file
- B.jpg → Cover image file

2. Open Command Prompt

- Press Windows + R.
- Type cmd and press Enter.
- Navigate to the folder using the cd command.

3. Check the Files

- List the files in the folder using: dir
- The folder should contain A.txt and B.jpg.

4. Hide the Text File Inside the Image File

- Use the following command: copy /b B.jpg + A.txt C.jpg
- This combines B.jpg and A.txt and creates a new file called C.jpg.

5. Verify the Output File

- Use: dir
- Check that C.jpg has been created.
- C.jpg now contains the image (B.jpg) + hidden text (A.txt).

6. Open C.jpg (Optional)

- Open C.jpg using any image viewer.
- The image will display normally even though the text file is appended to it.

PART 2 – HOW TO EXTRACT THE HIDDEN FILE

7. Extract the Hidden File

- Use the following command: copy /b C.jpg A_extracted.txt
- This extracts the hidden content into A_extracted.txt.
- It is better to use a different filename to avoid overwriting the original file.

8. Check the Extracted File

- Use: dir
- Verify that A_extracted.txt has been created.

9. View the Extracted Text

- Open A_extracted.txt using Notepad.
- Check the hidden content.
- Hidden text extracted successfully.

10. Clean Up (Optional)

- Remove the extracted file if it is no longer required.
- Keep the original evidence files unchanged during forensic examination.

OBSERVATION:

Secret Message: This is a secret message !!

SCREENSHOTS:

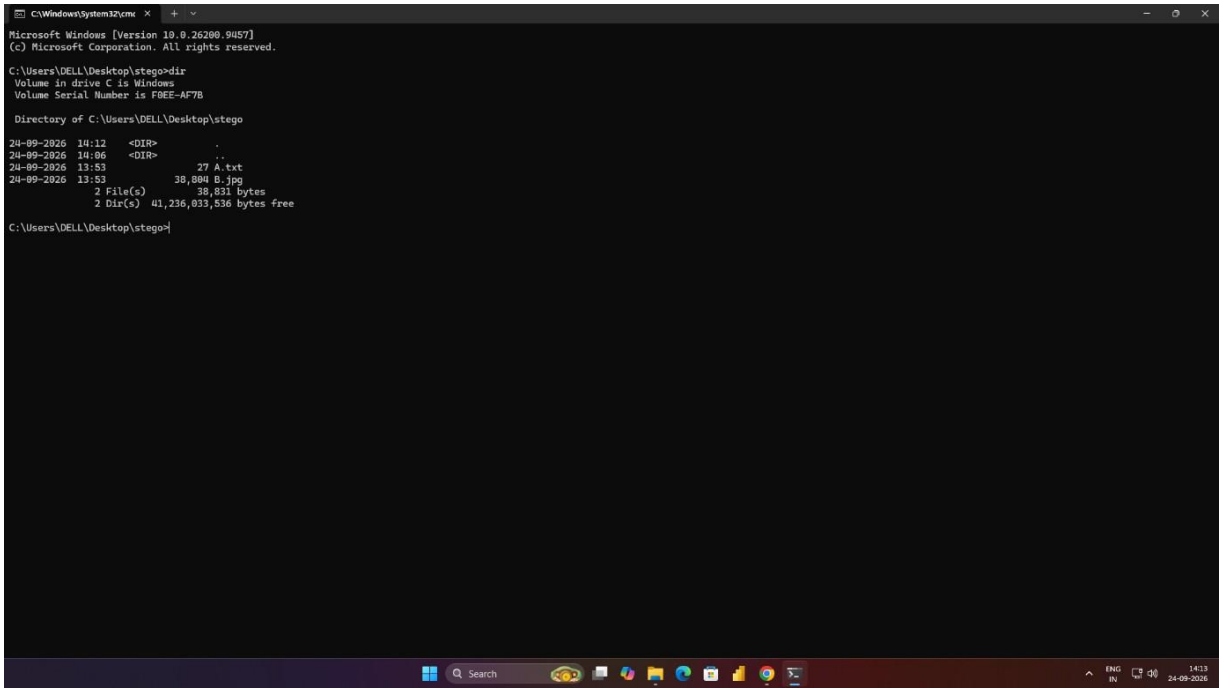


Figure 11.1 Display of 2 files in directory

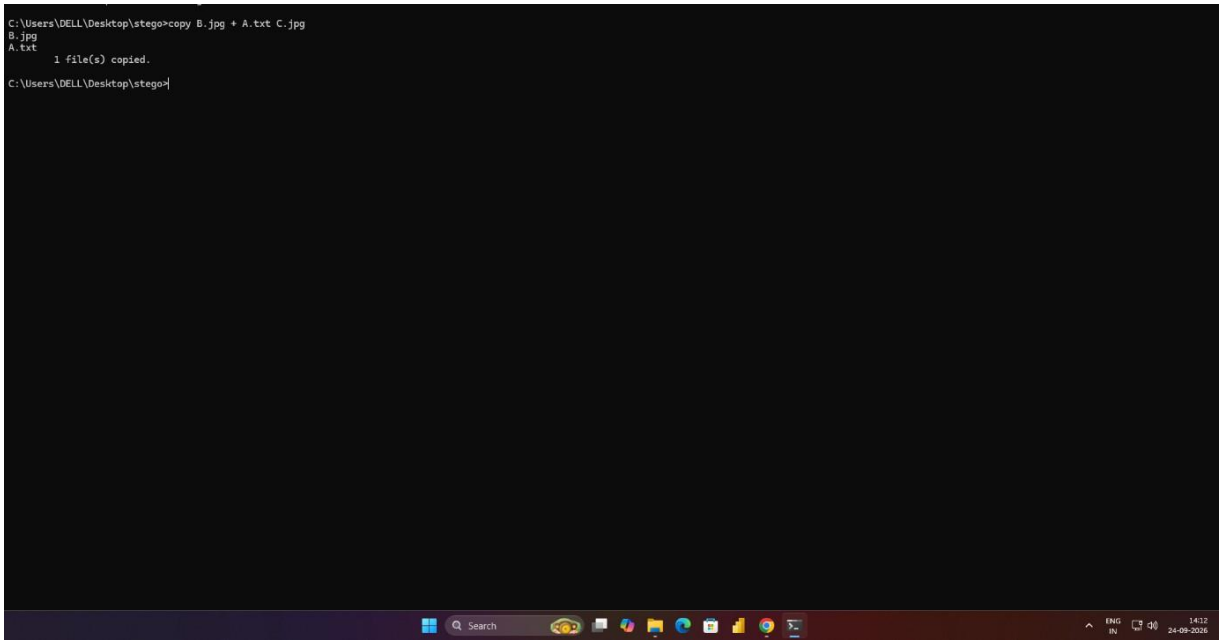


Figure 11.2 Hide txt file inside image file

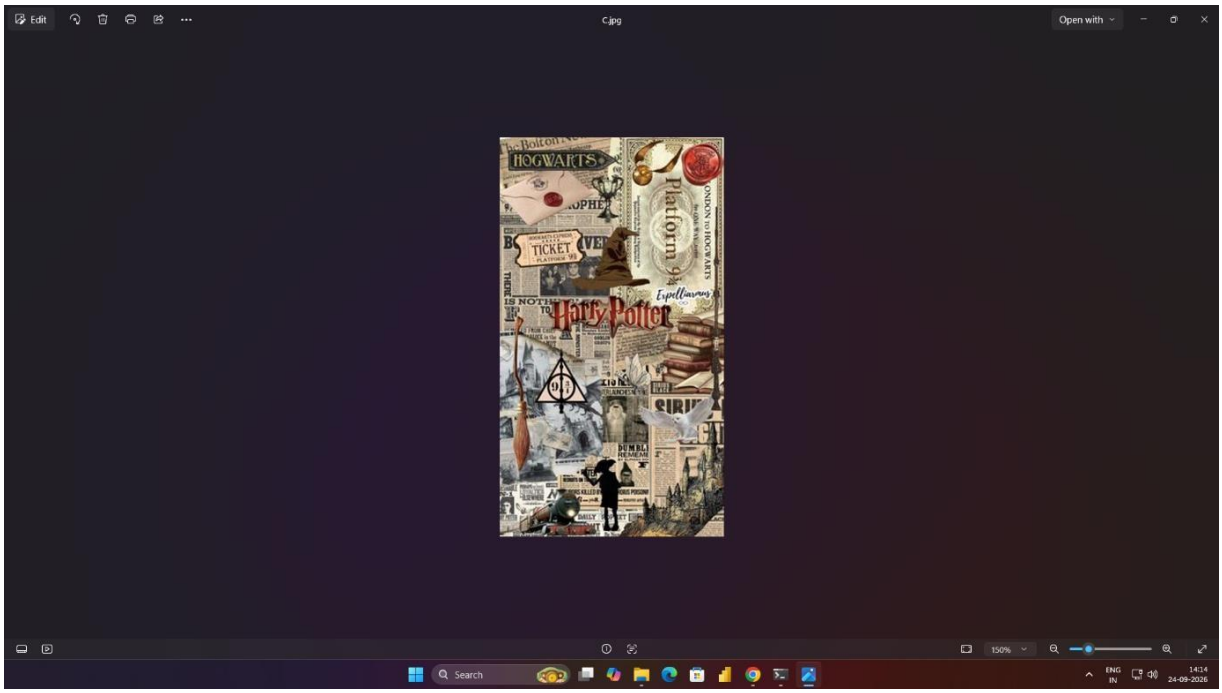


Figure 11.3 C.jpg

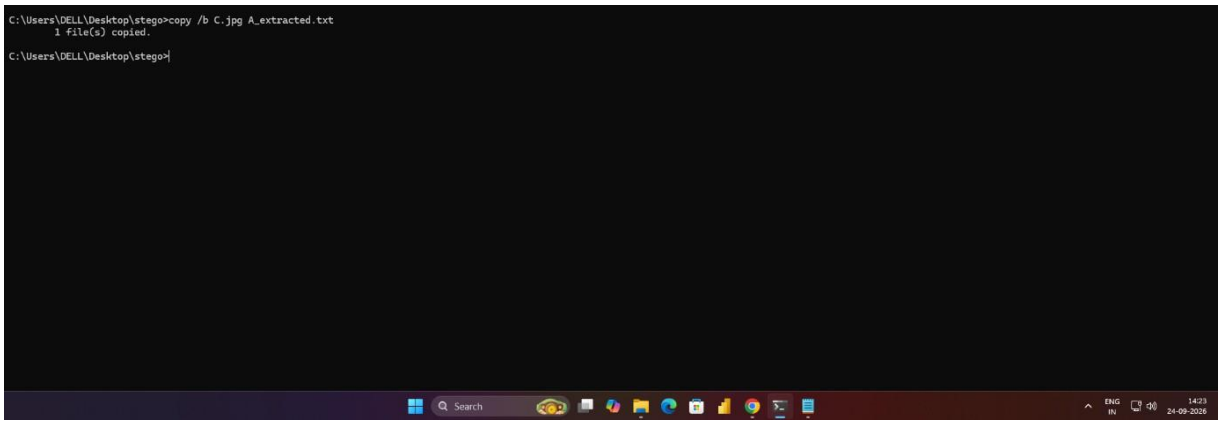


Figure 11.4 Extract C.jpg

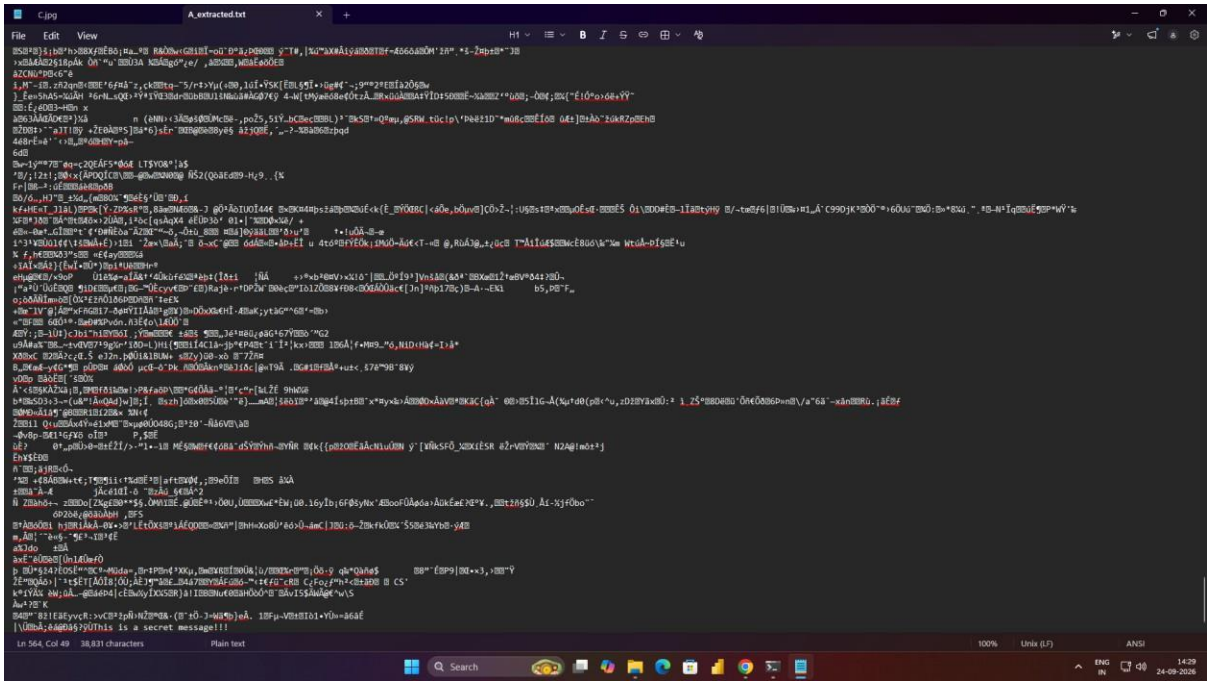


Figure 11.5 View Secret Message

RESULT

Thus a text file was appended to an image file, allowing the image to appear normal while additional data exists within the file.

EXPERIMENT NO: 12 **DATE: 01/10/2026**

HOW TO EXTRACT EXCHANGEABLE IMAGE FILE FORMAT (EXIF) DATA FROM IMAGE FILES USING EXIFREADER SOFTWARE

AIM

To extract Exchangeable Image File Format (EXIF) data from image files using ExifReader software.

TOOL

ExifReader (Free Software)

INTRODUCTION

Digital cameras and mobile devices store hidden information inside image files called EXIF data. This data may include camera model, date/time, GPS location, device make and model, and other information. ExifReader is a free Windows tool that allows users to view and analyze valuable forensic information from images.

PROCEDURE

1. Download ExifReader

- Open your browser and go to the official ExifReader website.
- Download the latest version of ExifReader.

2. Install ExifReader

- Run the downloaded setup file ExifReader.exe.
- Follow the installation wizard.
- Click Next and then Install.
- Click Finish to complete the installation.

3. Launch ExifReader

- Double-click the ExifReader icon.
- The main window of ExifReader will open.

4. Open an Image File

- Click on the File menu.
- Select Open Image File.
- Browse and select the image file, such as sample.jpg.
- Click Open.

5. View EXIF Data

- Once the image is loaded, ExifReader displays the EXIF metadata of the image.
- Scroll down to view complete information.
- Common EXIF information includes:

    1. Date/Time – When the photo was taken.
    2. GPS Location – Where the photo was taken.
    3. Camera Make & Model – Device used to capture the image.
    4. Exposure Settings – Camera settings used.
    5. Software – Editing software, if any, used on the image.

6. Save EXIF Report (Optional)

- Click on the File menu.
- Select Save Report.
- Choose the desired location.
- Save the report as a TXT or HTML file.

7. Analyze the EXIF Data

- Analyze important information such as:
    1. Date/Time – When the photo was taken.
    2. GPS Location – Where it was taken.
    3. Camera Make & Model – Device used.
    4. Exposure Settings – Camera settings.
    5. Software – Editing software, if any.

8. Document the Findings

- Note down the important EXIF details for the image.

- Attach the saved EXIF report to the case documentation.

9. Close the Application

- Close ExifReader after completing the analysis.
- Ensure all reports are saved.

OBSERVATION:

Image Width: 1292

Image Height: 503

Bits Per Sample: 8

Color Type: True Color with Alpha

Compression Type: Deflate

Filter Method: Adaptive

Interlace Method: No Interlace

sRGB Rendering Intent: Perceptual

Image Gamma: 0.455

Pixels Per Unit X: 3779

Pixels Per Unit Y: 3779

Unit Specifier: Metre

Detected File Type Name: PNG

Detected File Type Long Name: Portable Network Graphics

Detected MIME Type: image/png

Expected File Name Extension: .png

File Name: Screenshot 2025-09-03 100543.png

File Size: 85154 bytes

File Modified Date: Thu Sep 03 10:05:43 +05:30 2025

SCREENSHOTS:

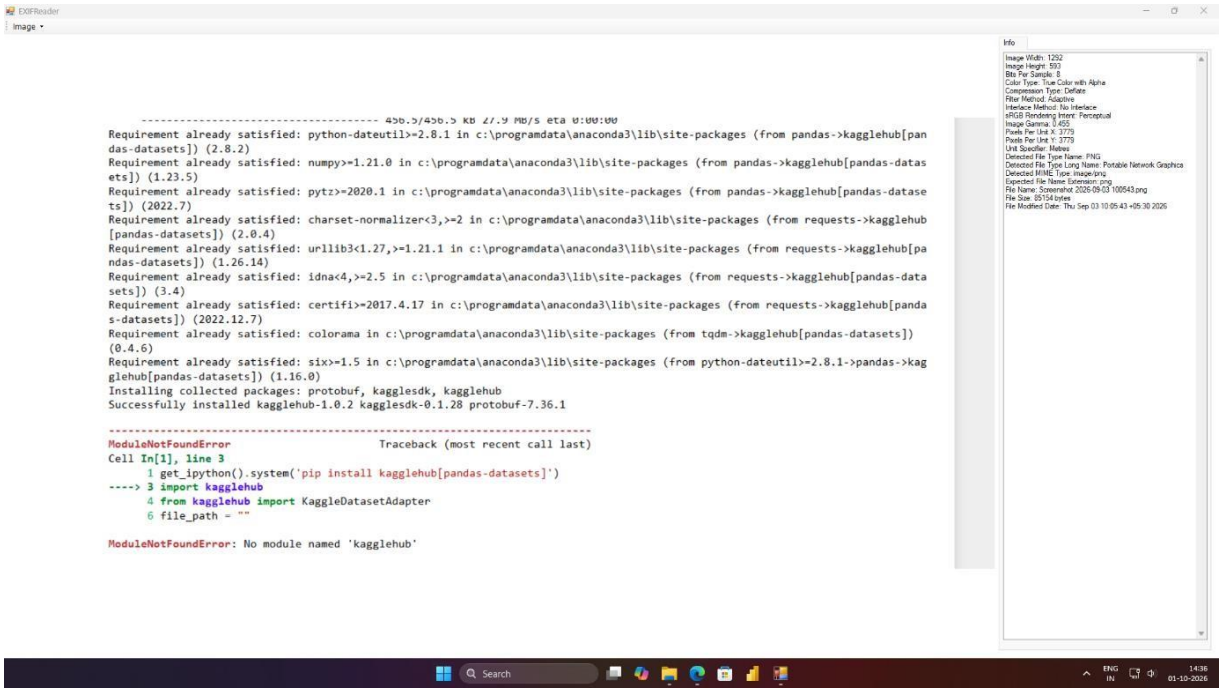


Figure 12.1 ExifReader displays metadata of the image

RESULT

Thus ExifReader was used to find EXIF data about the image and the findings were documented.